Pelvic Viscera I

JeffMD 2026

EUR Block

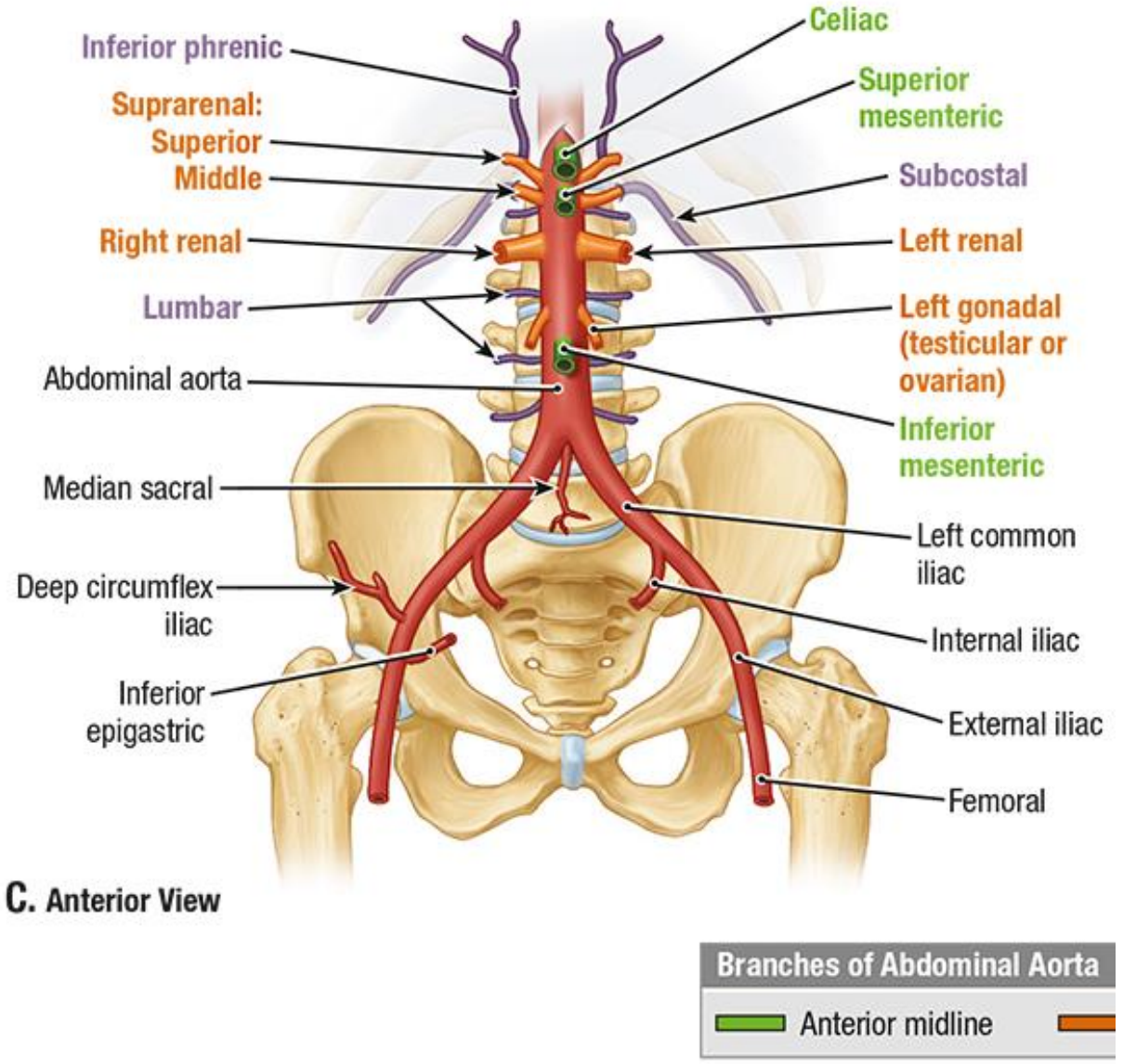
Elizabeth A Spudich, PhD

Learning Objectives Viscera I

- Discuss the vasculature supply to the pelvis and perineum, including relationships and anastomoses with the systemic and portal systems
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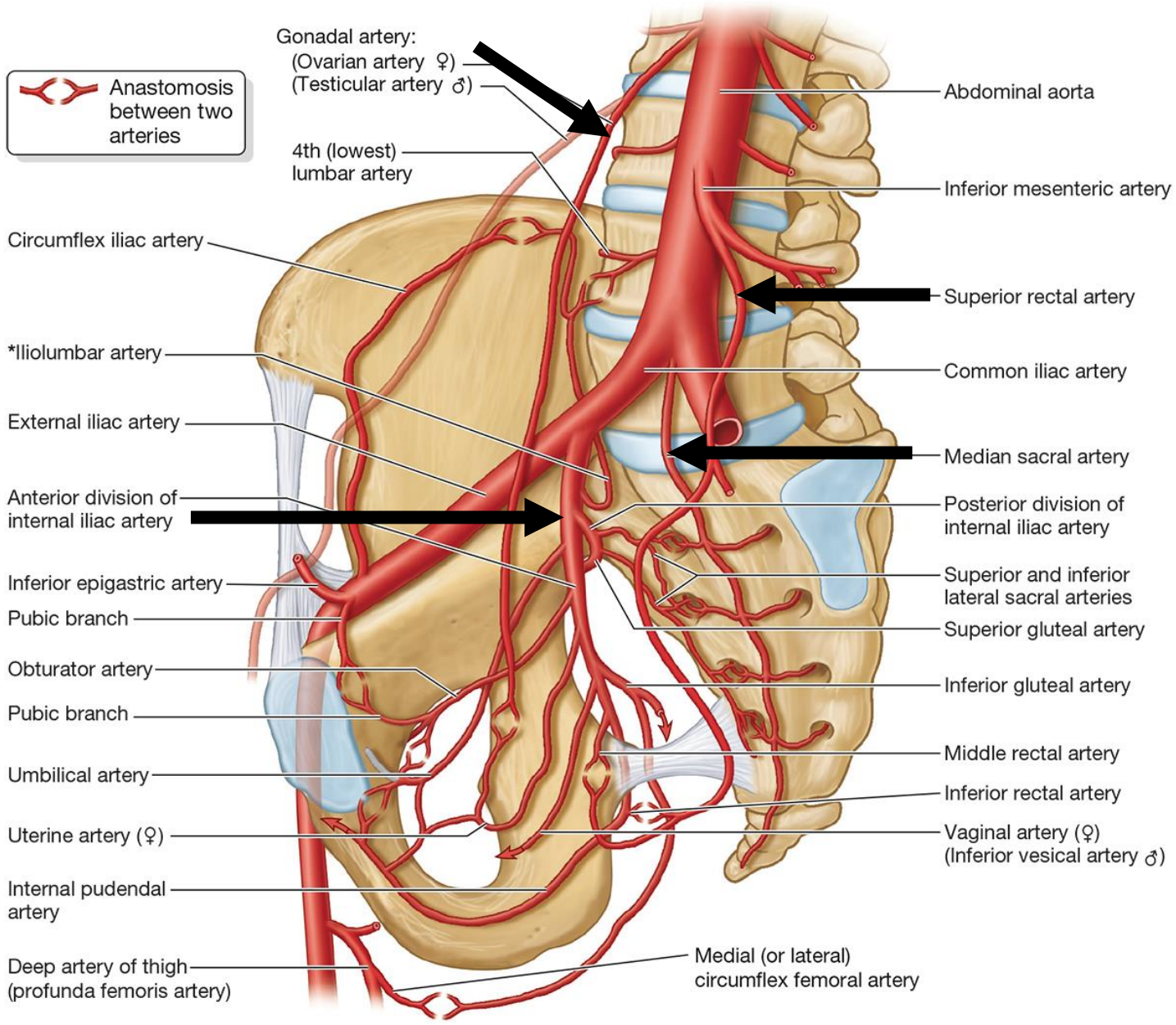
Arteries of the True Pelvis

- Abdominal Aorta branches to form the Common Iliac
- Common will bifurcate to the internal and external iliac arteries



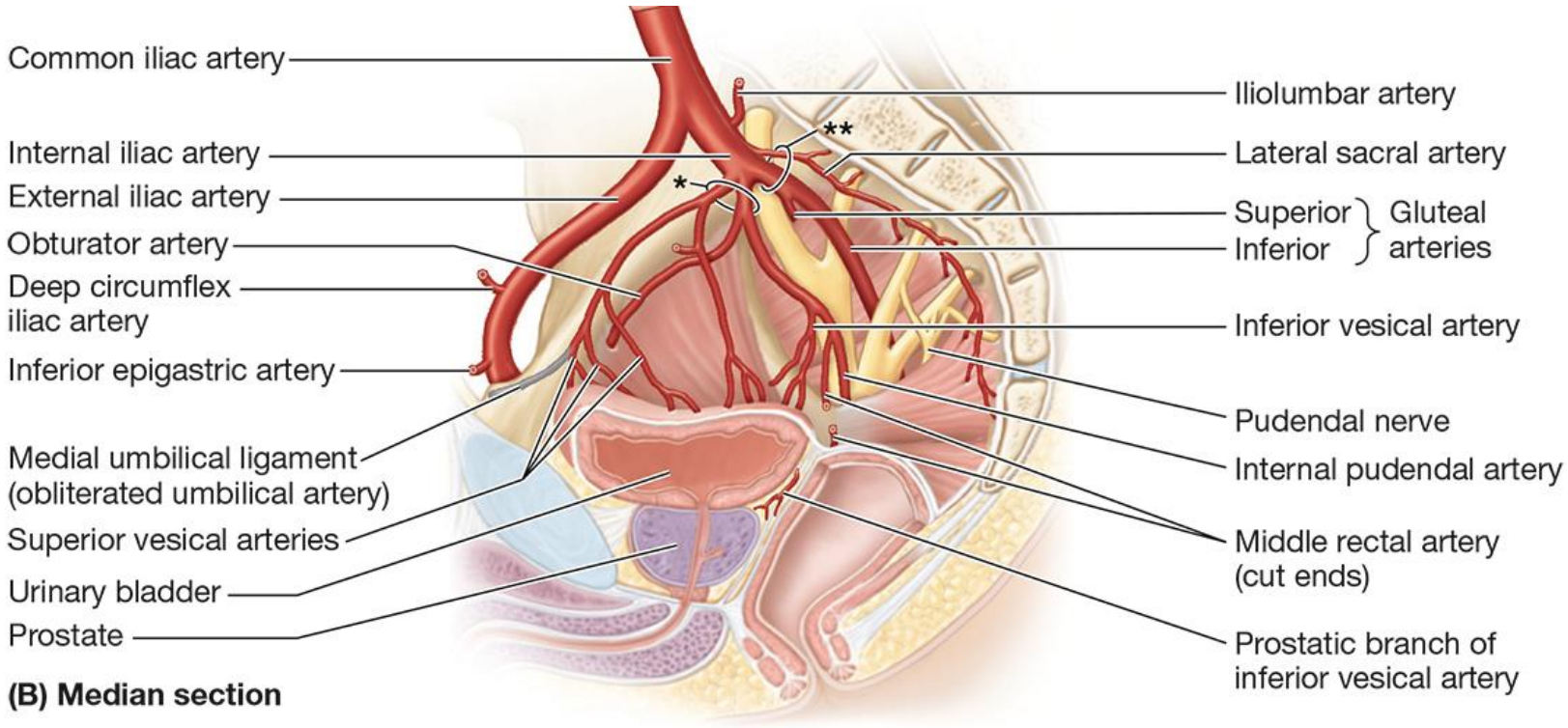
Arteries of the True Pelvis

- True Pelvis has 3-4 aortic tributaries
  - Internal Iliacs
  - Median Sacral
  - Superior Rectal (end of inferior mesenteric)
  - Ovarian (if present)



Internal Iliac Artery

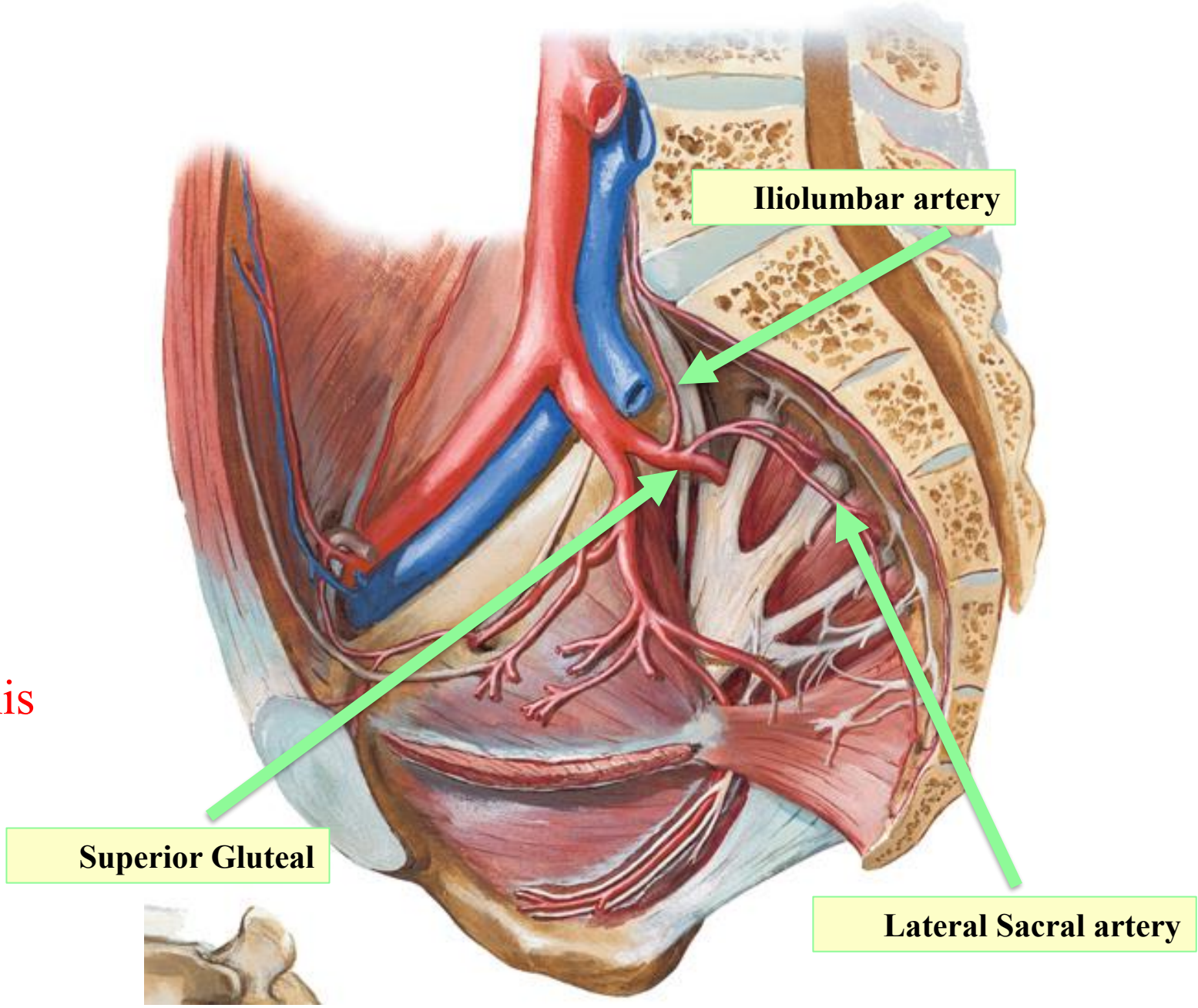
- Anterior branch supplies organs
- Posterior branch Supplies muscles and walls
- Variation is the rule
  - 9 basic patterns
  - 49 sub patterns



I don't care where it comes from I only care where it goes

Posterior
Internal Iliac Branches

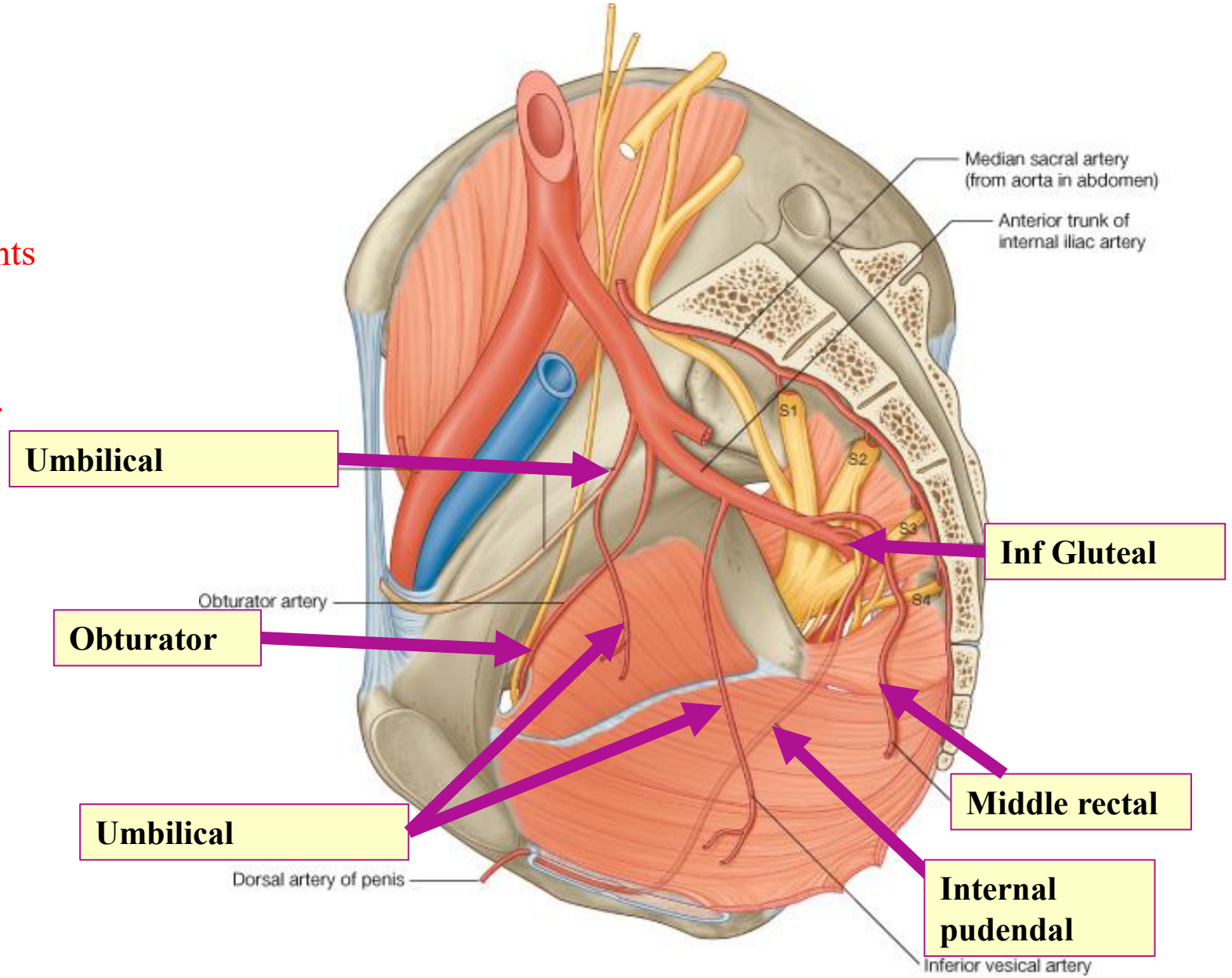
- Iliolumbar artery
  - Ascends to lumbar region
- Lateral sacral artery
  - Supplies the lateral mass of sacrum
- Superior Gluteal artery
  - Exits superior to the piriformis
  - Through the greater sciatic foramen
  - Usually between the lumbosacral trunk and S1 ventral root



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Anterior Internal Iliac Branches

- Umbilical artery
  - Becomes the medial umbilical ligaments in wall
- Obturator Artery
  - Picks up the nerve and exits obturator foramen
- Superior and inferior vesicular
- Internal pudendal
  - out greater; back in to the perineum through lesser
- Middle rectal
- Inferior gluteal
  - Distal end of Anterior Iliac
  - Exits inferior to piriformis muscle

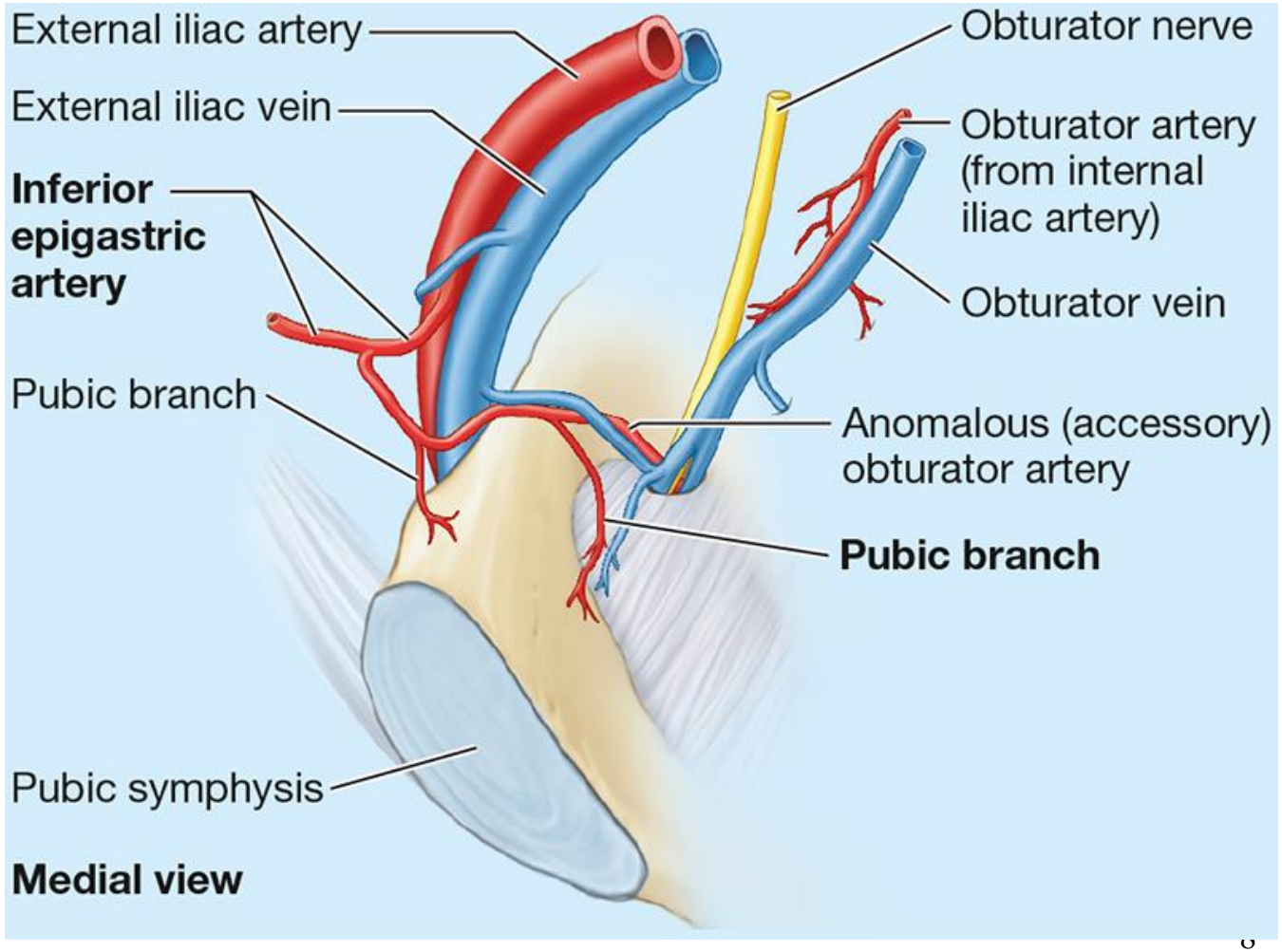


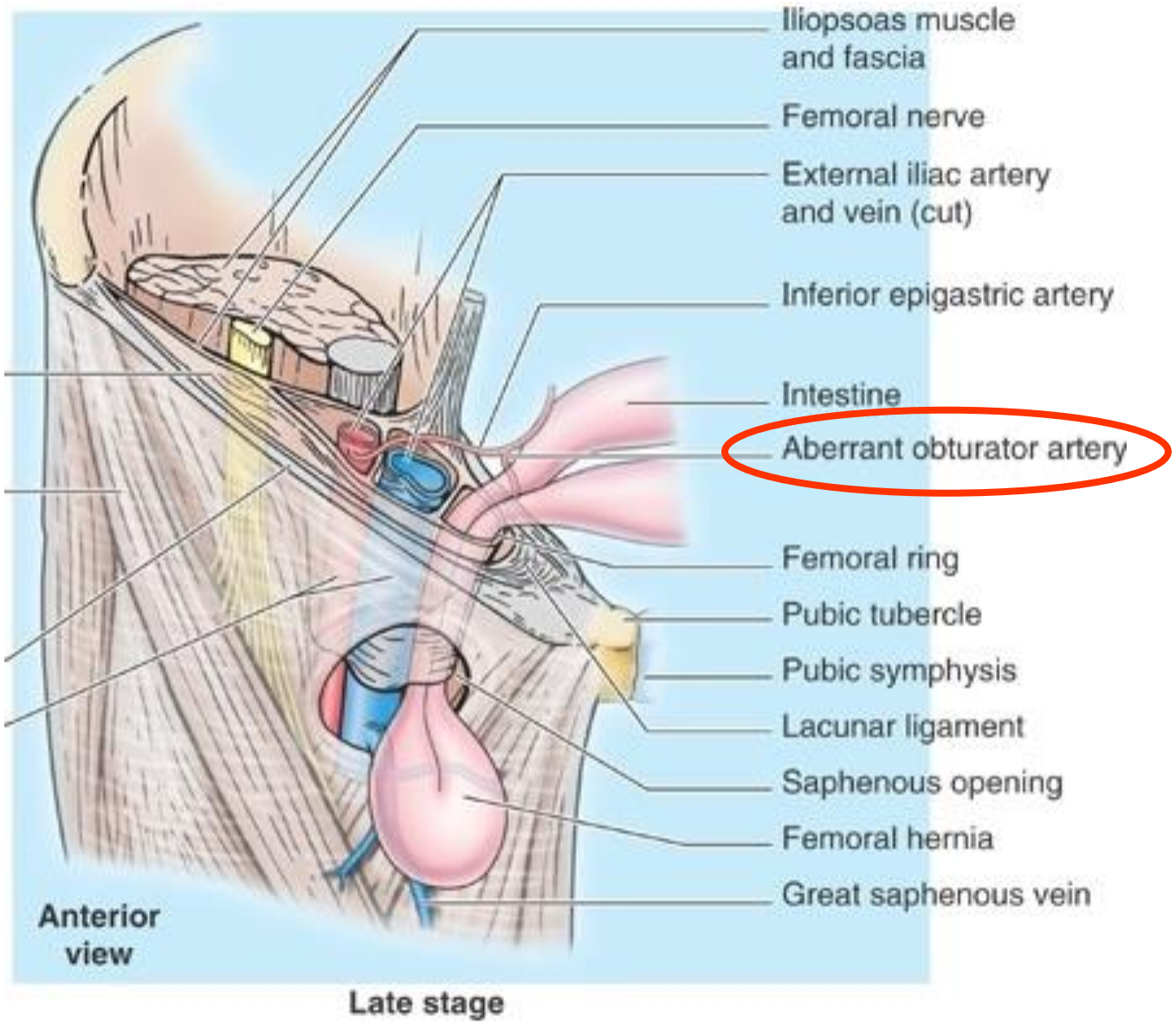
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Accessory (Anomalous) Obturator Artery 20%

A large branch from the external iliac or usually inferior epigastric a. is present and obturator a. from internal iliac a. could be very small or absent.

Clinical significance: may be lacerated during reduction or repair of a femoral hernia. Surgeons must be aware the possible presence of this common arterial variant.





Close to or crosses the femoral ring

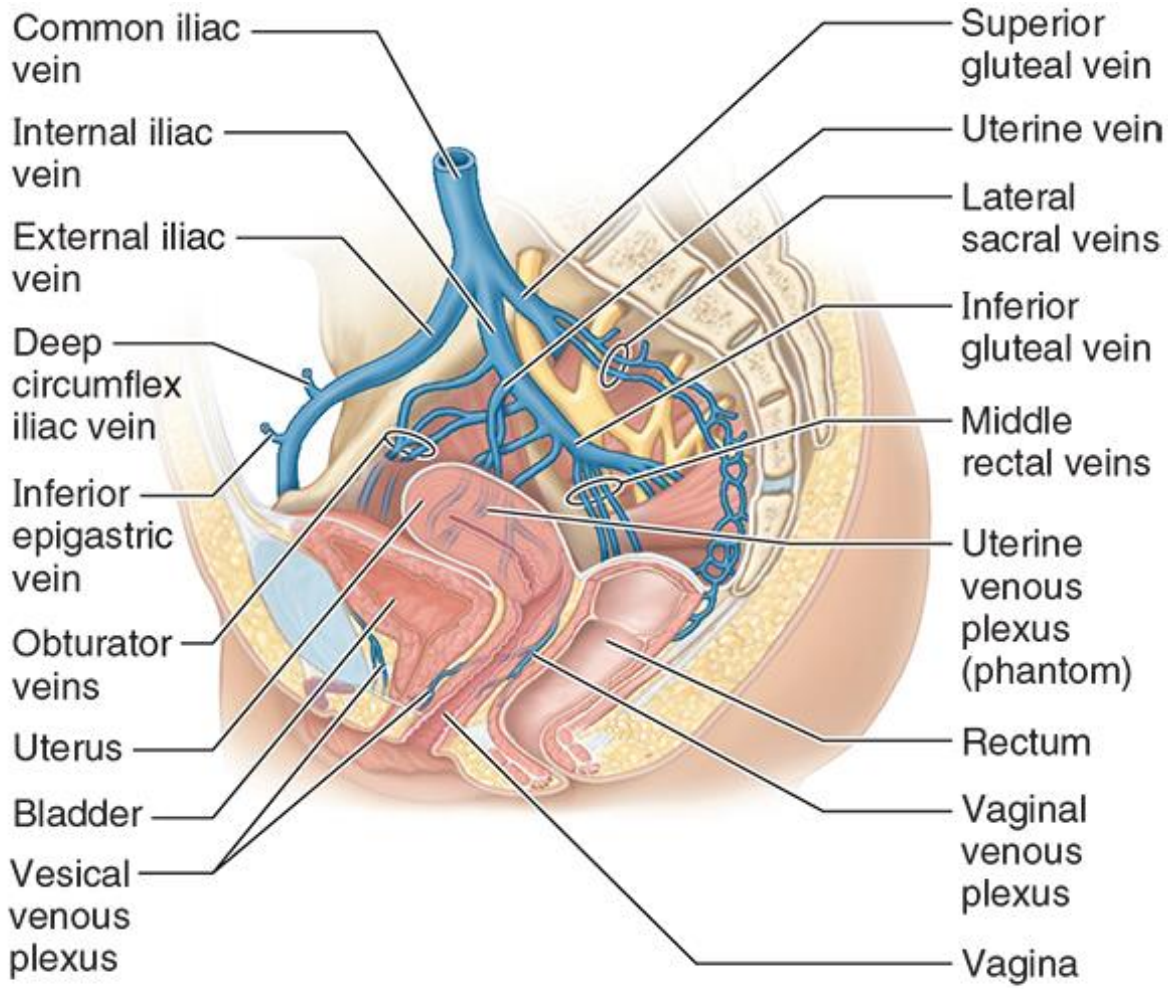
Closely related to the neck of a femoral hernia

Clinical significance: may be lacerated during reduction or repair of a femoral hernia. Surgeons must be aware the possible presence of this common arterial variant.

Pelvic Venous Drainage

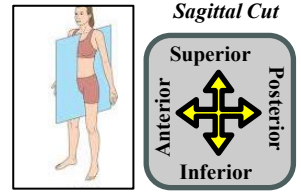
- Veins parallel arteries of the same name
- There are also ***venous plexuses*** surrounding the viscera
  - Prostatic
  - Uterine
  - Vaginal
  - Vesicle

Correlate: These intercommunicating networks of veins are clinically important, as they may serve as a route for spread of infection or metastatic cells



(B) Median section

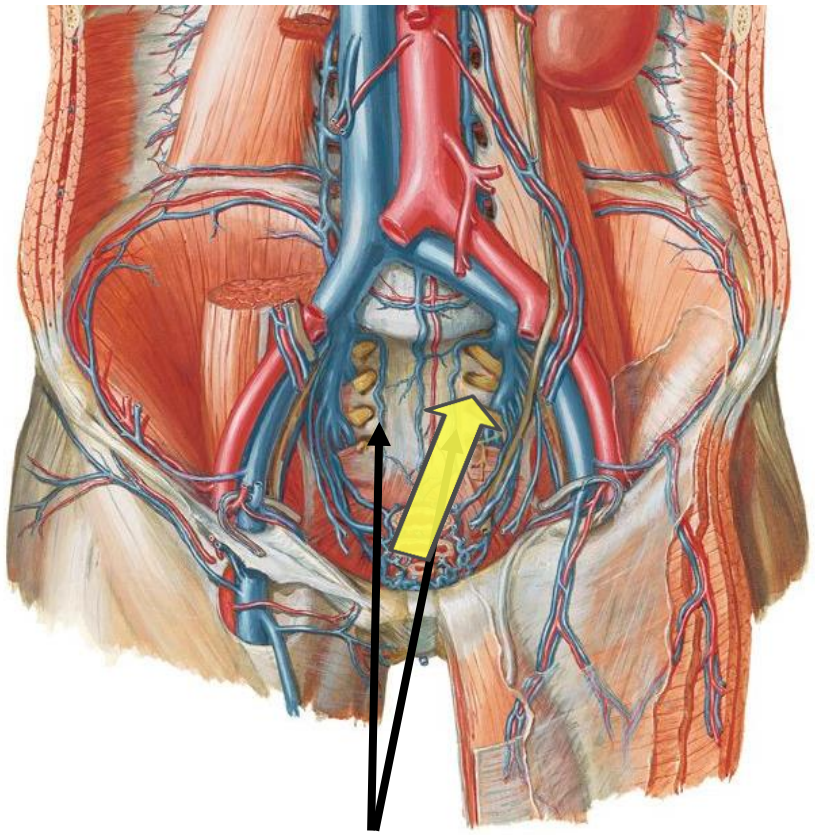
Medial View, Sagittal Cut



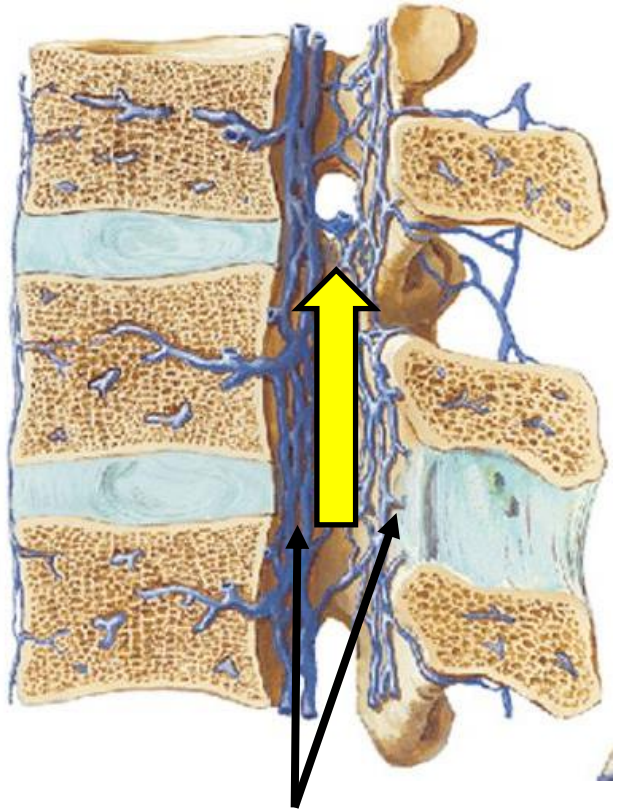
Moore's, 9th ed., Fig. 6.19

Slides adapted from Dr. David Rascici at WVU

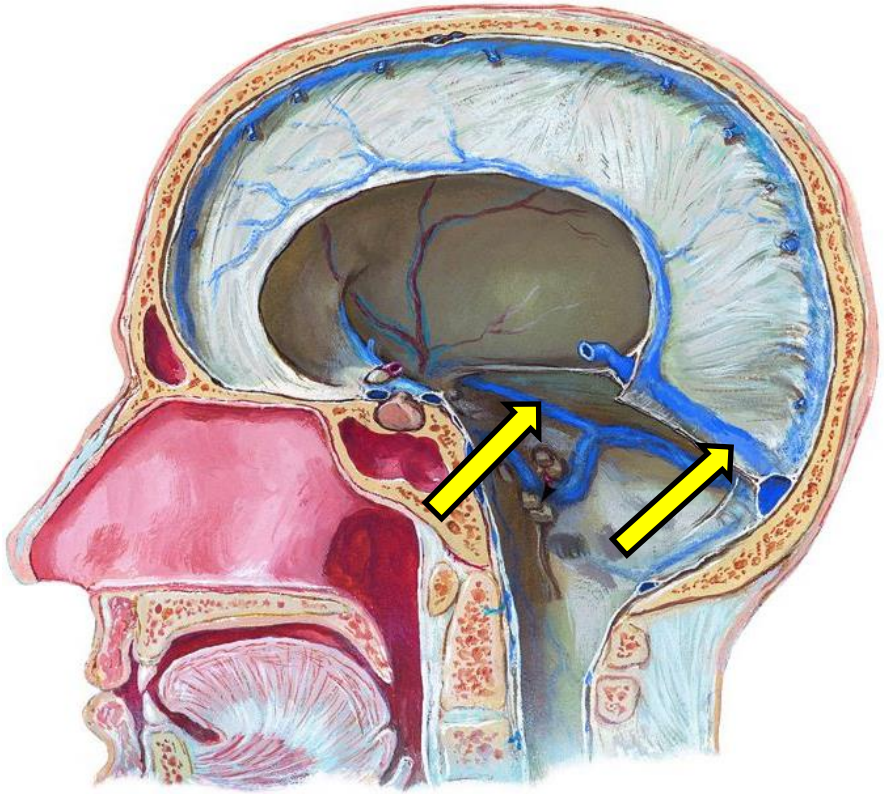
The ***lateral sacral veins*** provide access to the ***internal vertebral venous plexus (valveless)*** which can provide an anatomical account of metastatic spread from the pelvic cavity to the brain.



Lateral sacral veins (through anterior sacral foramina)



Internal vertebral venous plexus



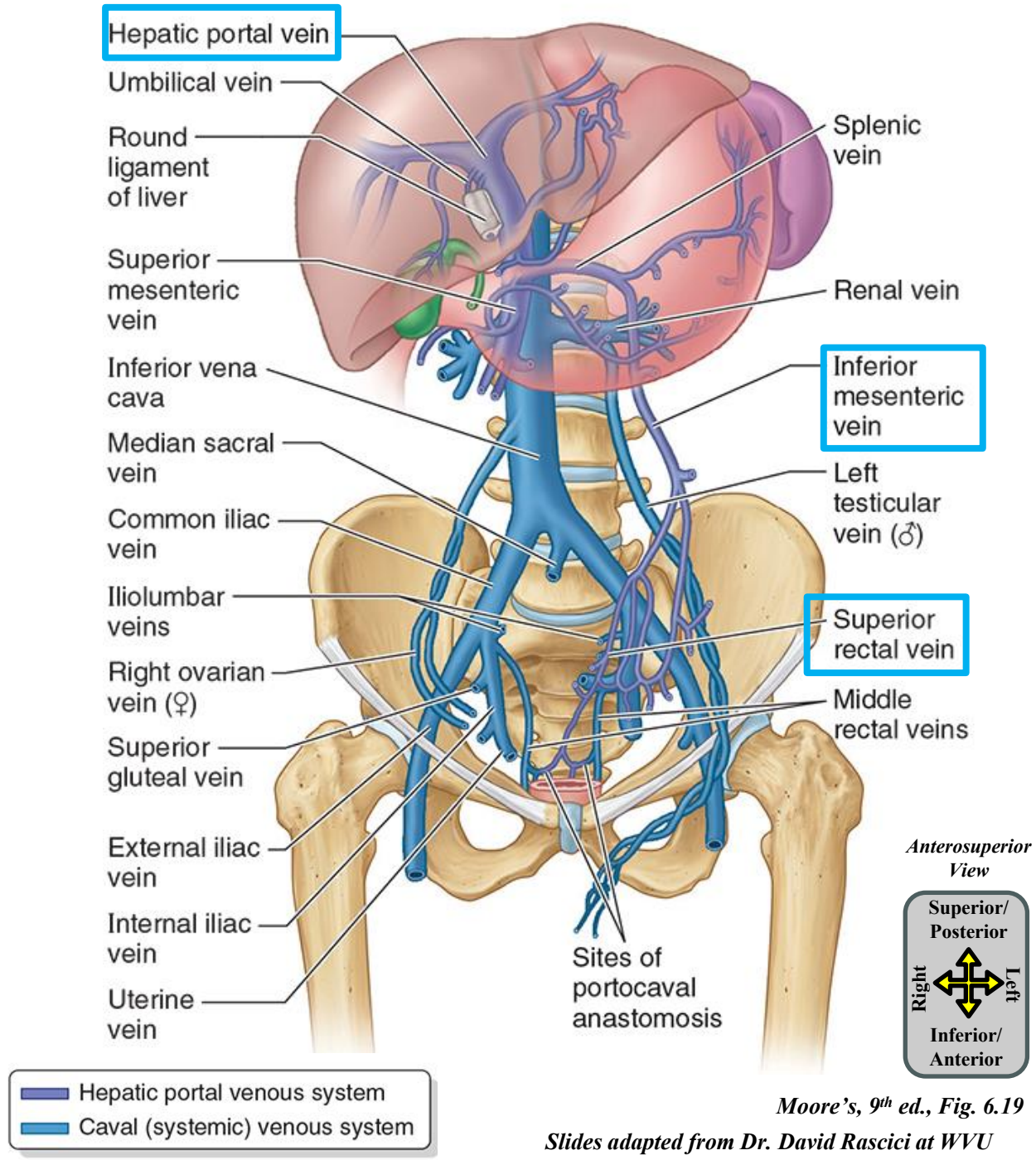
Dural Venous Sinuses & Brain Tissue

Slides adapted from Dr. David Rascici at WVU

The Hepatic Portal and Caval Systems

Alternative routes of venous drainage include the hepatic portal venous system

- **Superior rectal vein** *drains into the* ***hepatic portal vein*** *via the inferior mesenteric vein*
- Anastomosis with the caval system (IVC)
  - Middle rectal
  - Inferior rectal



Moore's, 9th ed., Fig. 6.19

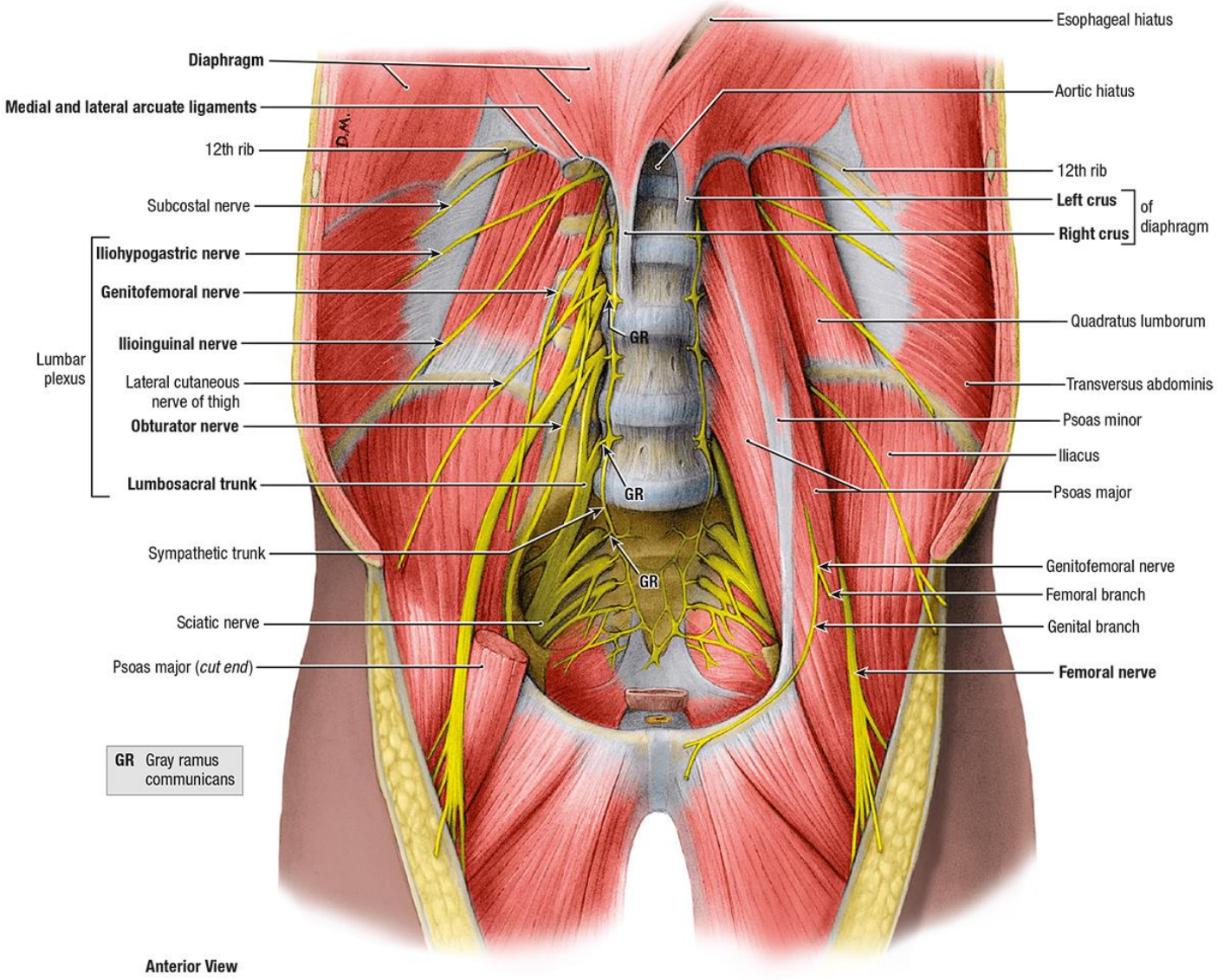
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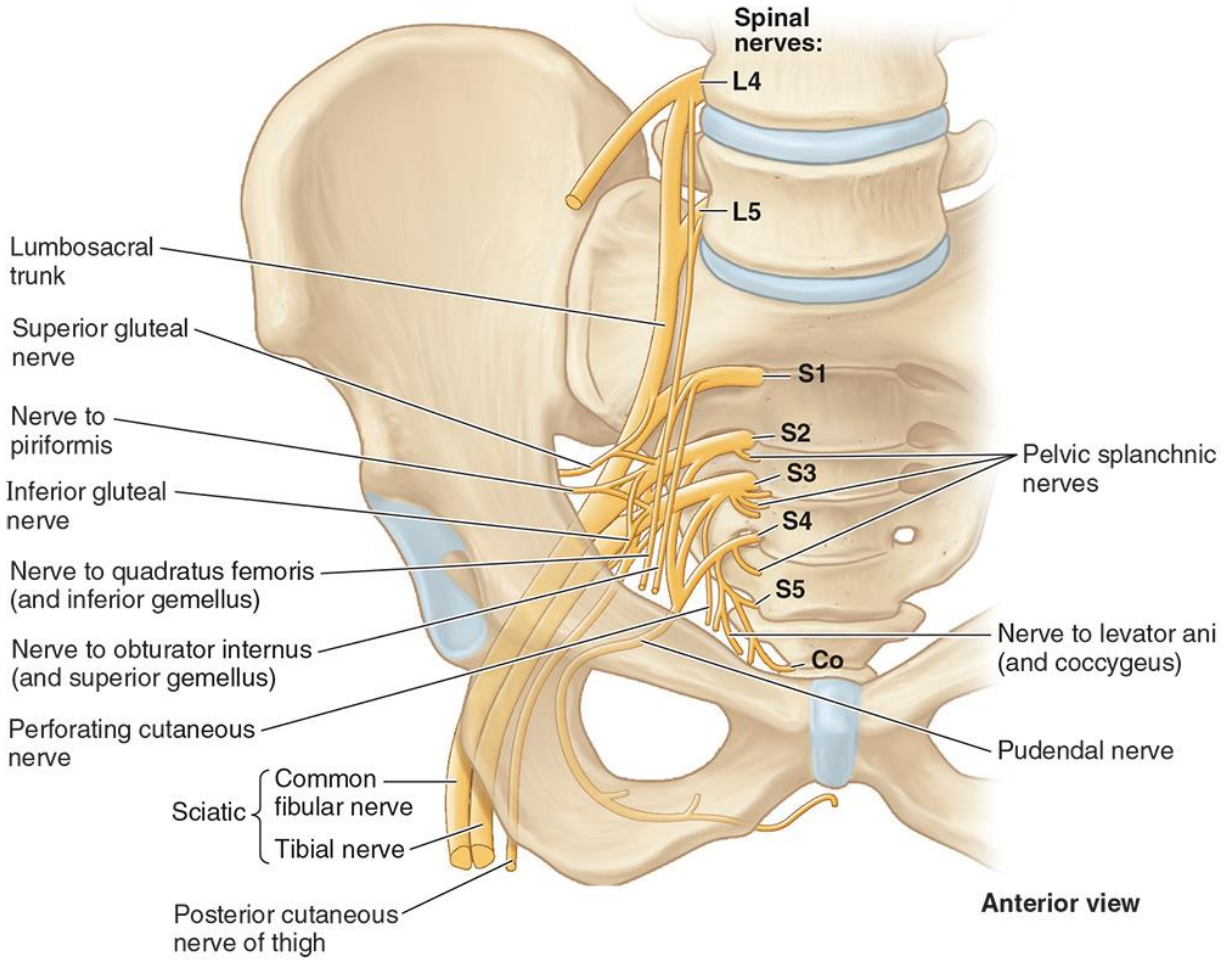
Lumbar Nerves of the Pelvis

- Lumbar plexus branches
  - Obturator (L2-4)
  - Lumbosacral trunk (L4-5)



Sacral Plexus (L4-S4)

- Motor and Sensory
- Superior Gluteal (L4-S1)
  - Glut medius and minimus
- Inferior Gluteal (L5-S2)
  - Glut Max
- Pudendal (S2-S4)
  - All of perineum
- Sciatic (L4-S3)
- Posterior cutaneous nerve to thigh(S1-S3)
  - Posterior thigh and leg

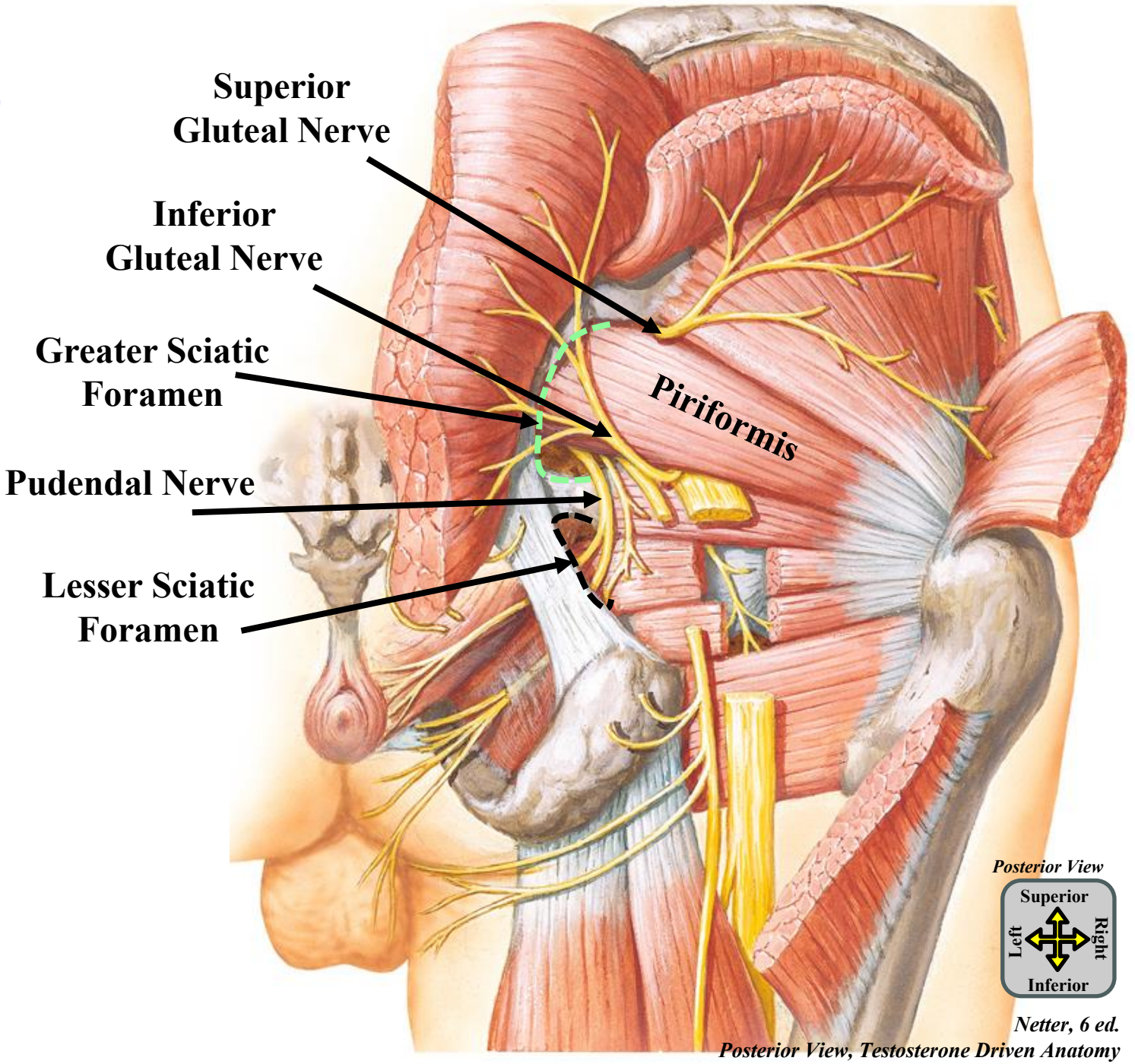


Sacral Plexus (L4-S4)

Sciatic nerve, inferior and superior gluteal nerves, and pudendal nerve **ALL** exit the pelvic cavity via the **greater sciatic foramen**

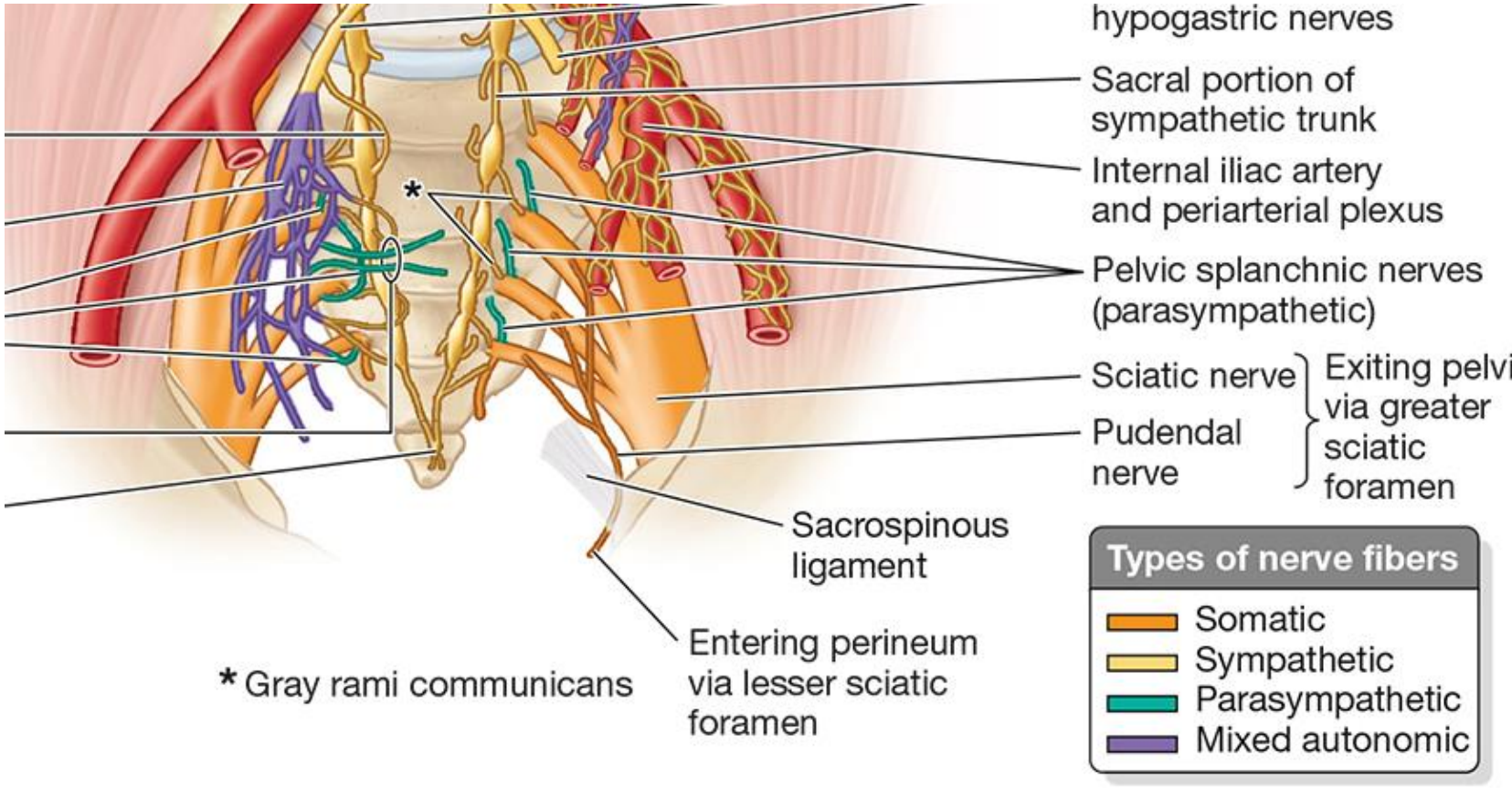
Inferior gluteal nerve exits inferior to the piriformis muscle, whereas superior gluteal exits superior to the piriformis muscle

Pudendal nerve exits the gluteal region **through lesser sciatic foramen** to provide motor and sensory to the perineum



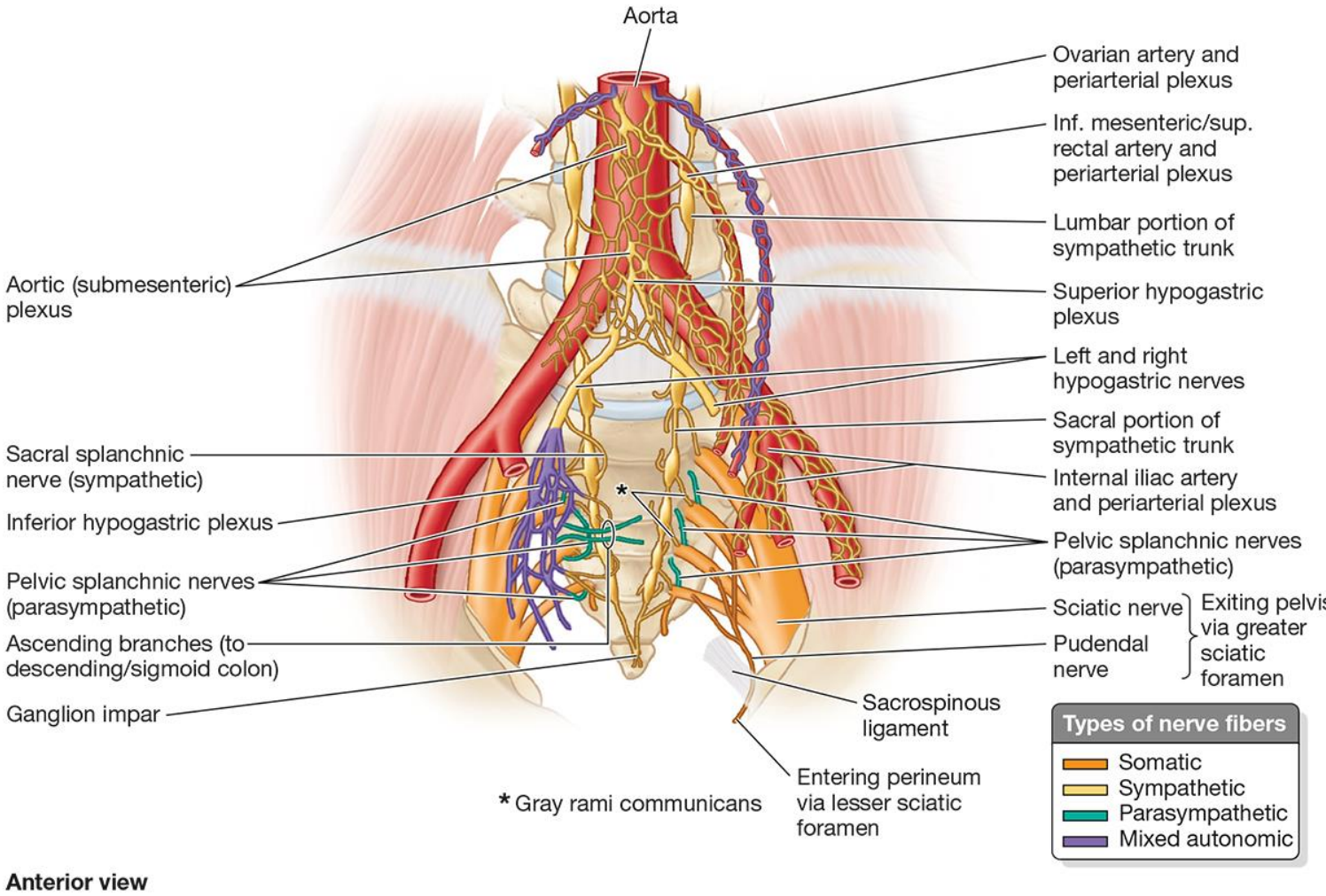
Pelvic Splanchnic Nerves

- Sacral nerves S2-S4
- Preganglionic Parasympathetic nerves
- Merges with the Sympathetic Postganglionic fibers in the inferior hypogastric plexus



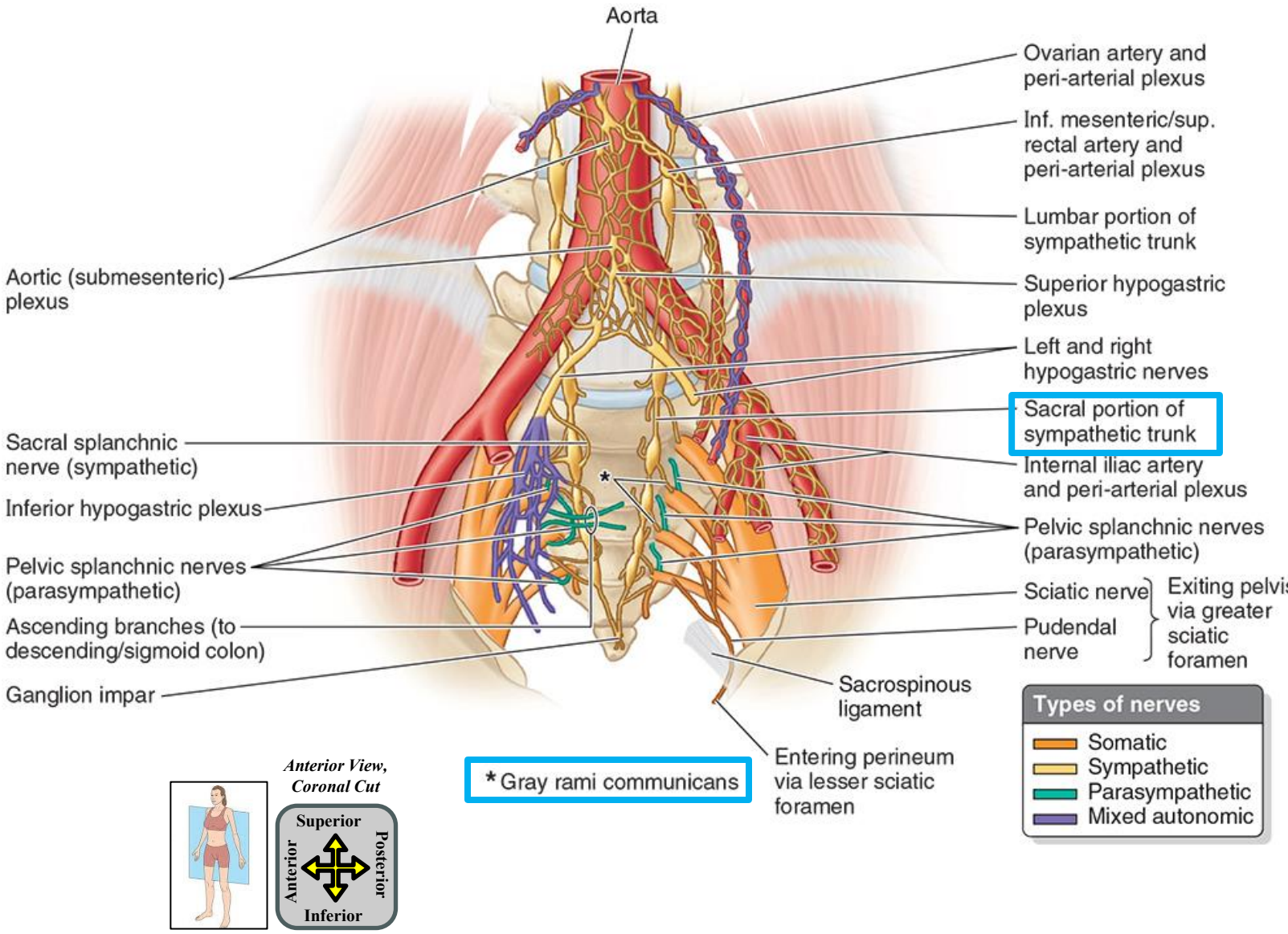
Sympathetic Innervation to the Pelvis

- Superior hypogastric plexus
  - Extension of mesenteric fibers
  - Motor and afferent fibers
  - Supplies ureters and gonads
- Hypogastric nerves
  - Branch down to enter the pelvis
- Inferior hypogastric plexuses
  - Postsynaptic sympathetics of hypogastric nerves
  - Presynaptic parasympathetic fibers of pelvic splanchnic
  - Visceral afferent



Sacral Sympathetic Trunk

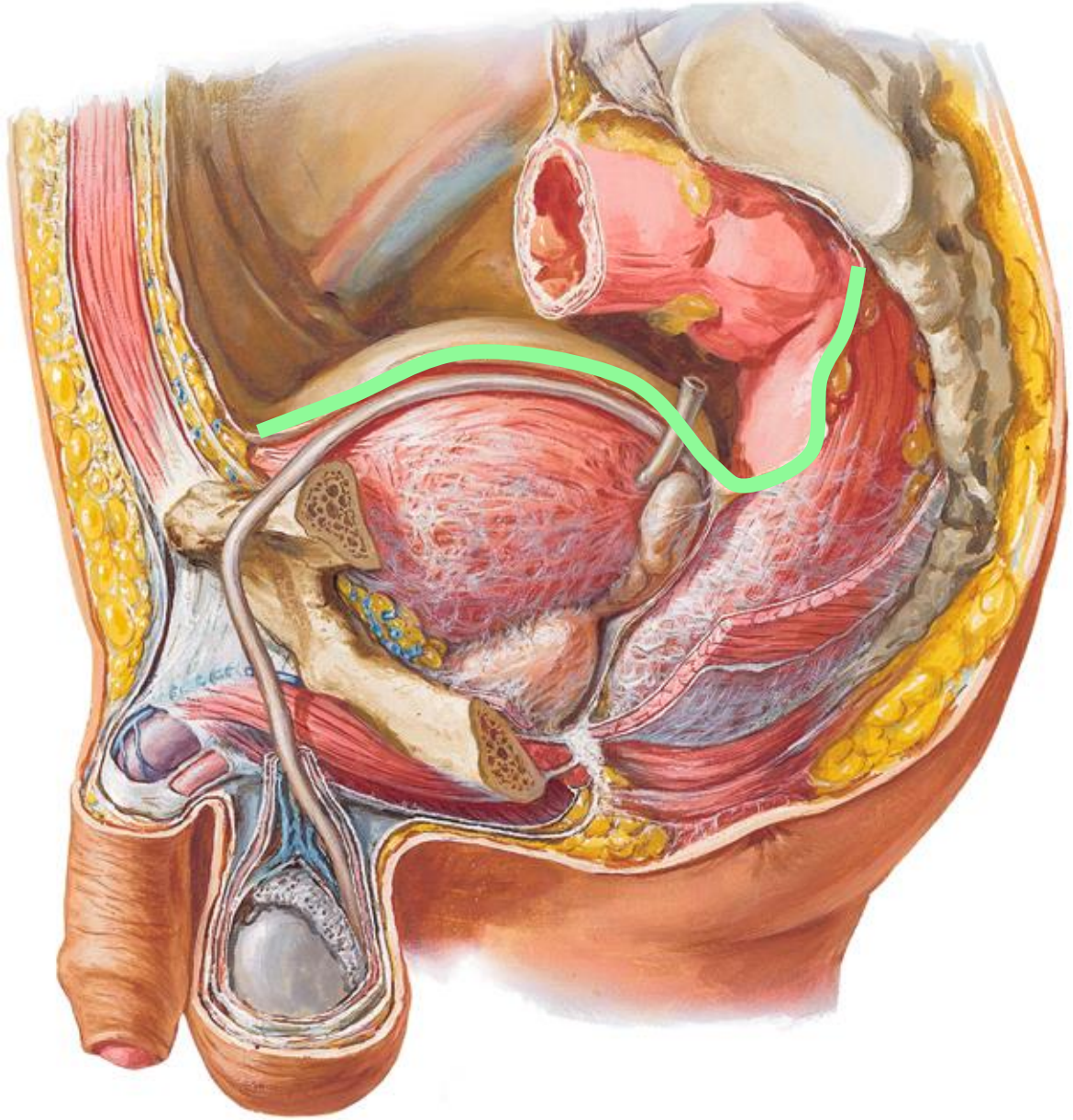
- Continuation of the lumbar sympathetic trunks
- Contains 4 ganglia medial to the sacral foramina
- Sends branches ***(sacral splanchnics)*** to the ***inferior hypogastric plexus***
- Function: postsynaptic sympathetic innervation of the lower limb
  - Vasomotor, pilomotor (hair on skin), sudomotor (sweat glands)



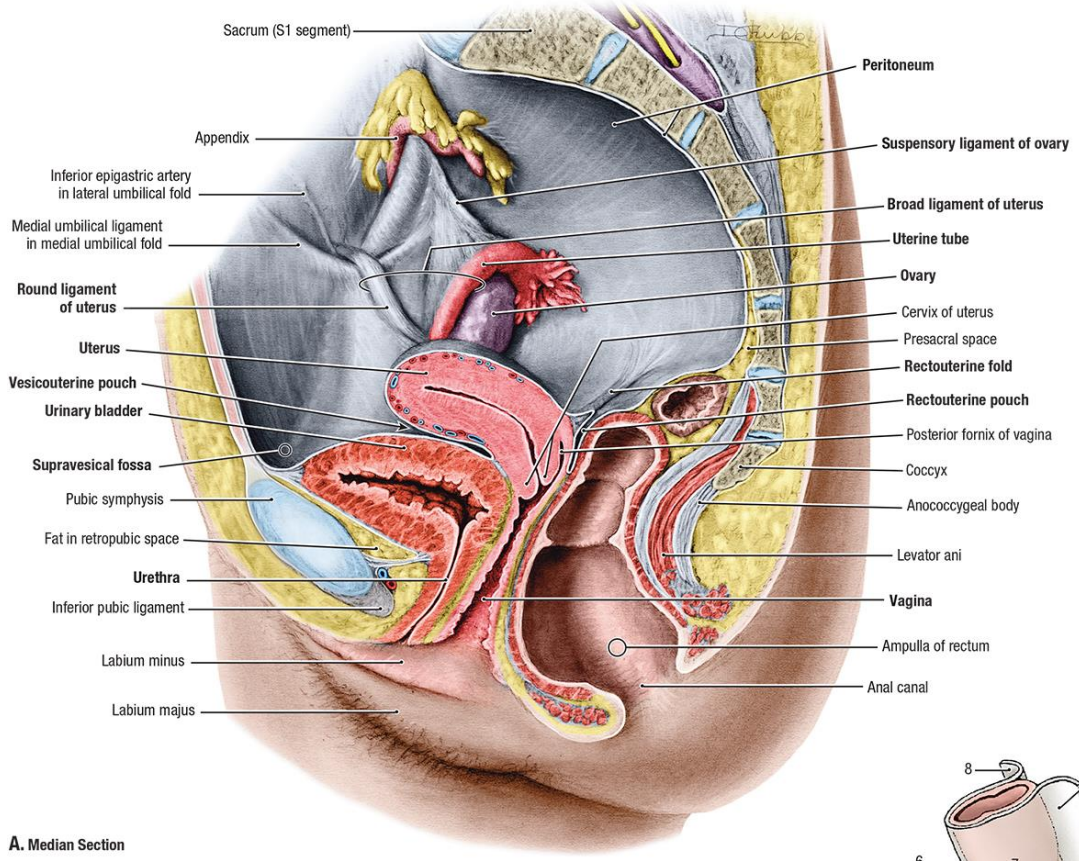
Moore's, 9th ed., Fig. 6.23

Pelvic Pain Line

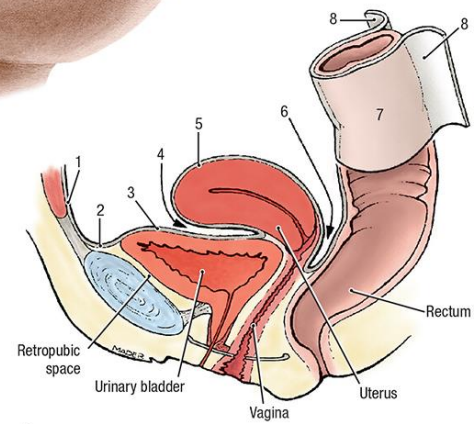
- Inferior limit to the peritoneum
- Pelvic organ regions covered with peritoneum send afferents to Spinal ganglia T11-L2/3
  - Follows sympathetic fibers
- Regions of pelvic organs NOT covered with peritoneum send afferents to spinal ganglia of S2-S4
  - Follows Parasympathetic fibers



Peritoneal Reflections/Pain Line

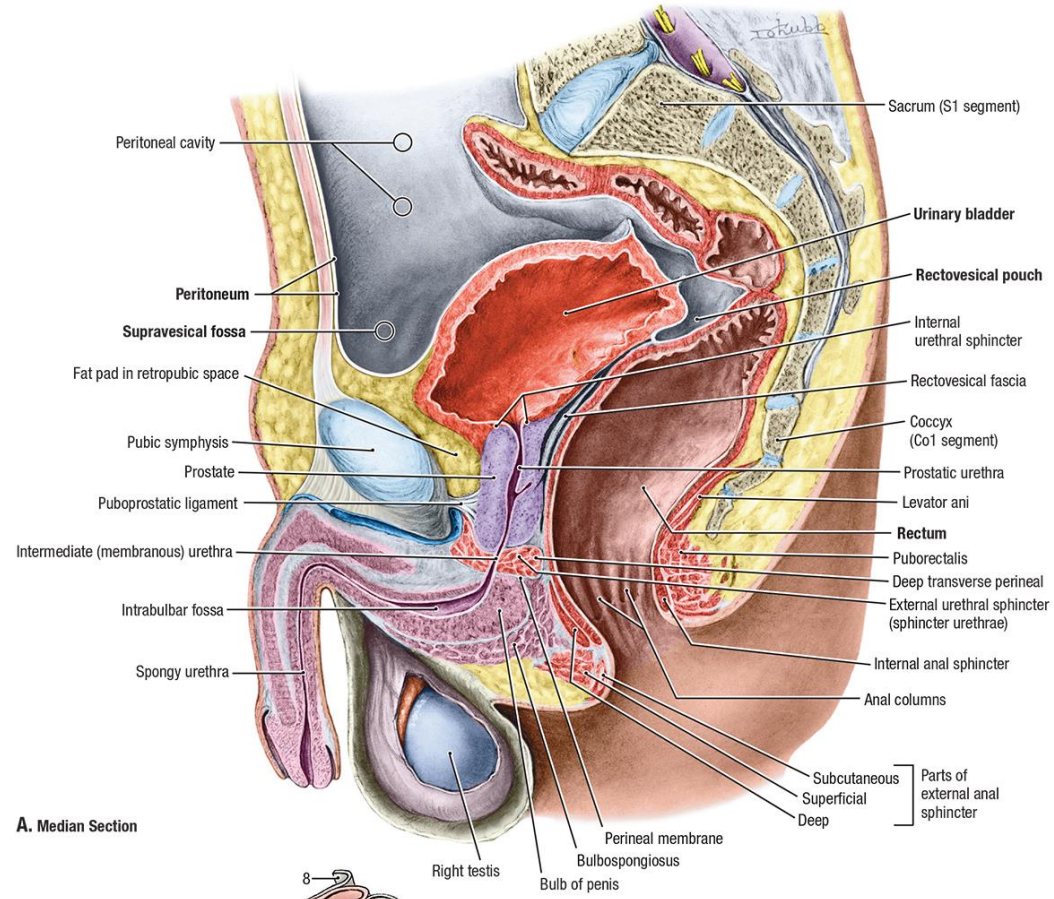


A. Median Section

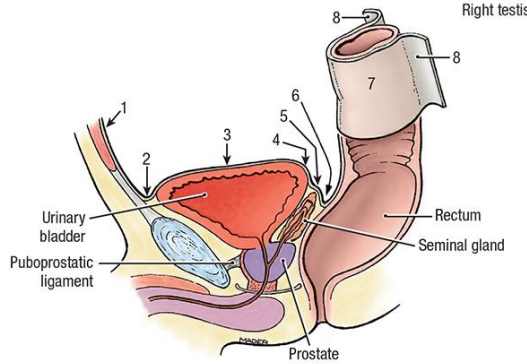


B. Schematic, Median Section

Peritoneal Reflections in Females (in Midline) in B
Peritoneum passes:
1 From the anterior abdominal wall
2 Superior to the pubic bone, forming supravesical fossa
3 On the superior surface of the urinary bladder
4 From the bladder to mid-uterus, forming the vesicouterine pouch
5 On the fundus and body of the uterus, and posterior fornix of the vagina
6 Between the rectum and uterus, forming the rectouterine pouch
7 On the anterior and lateral sides of the rectum
8 Posteriorly to become the sigmoid mesocolon



A. Median Section



B. Schematic, Median Section

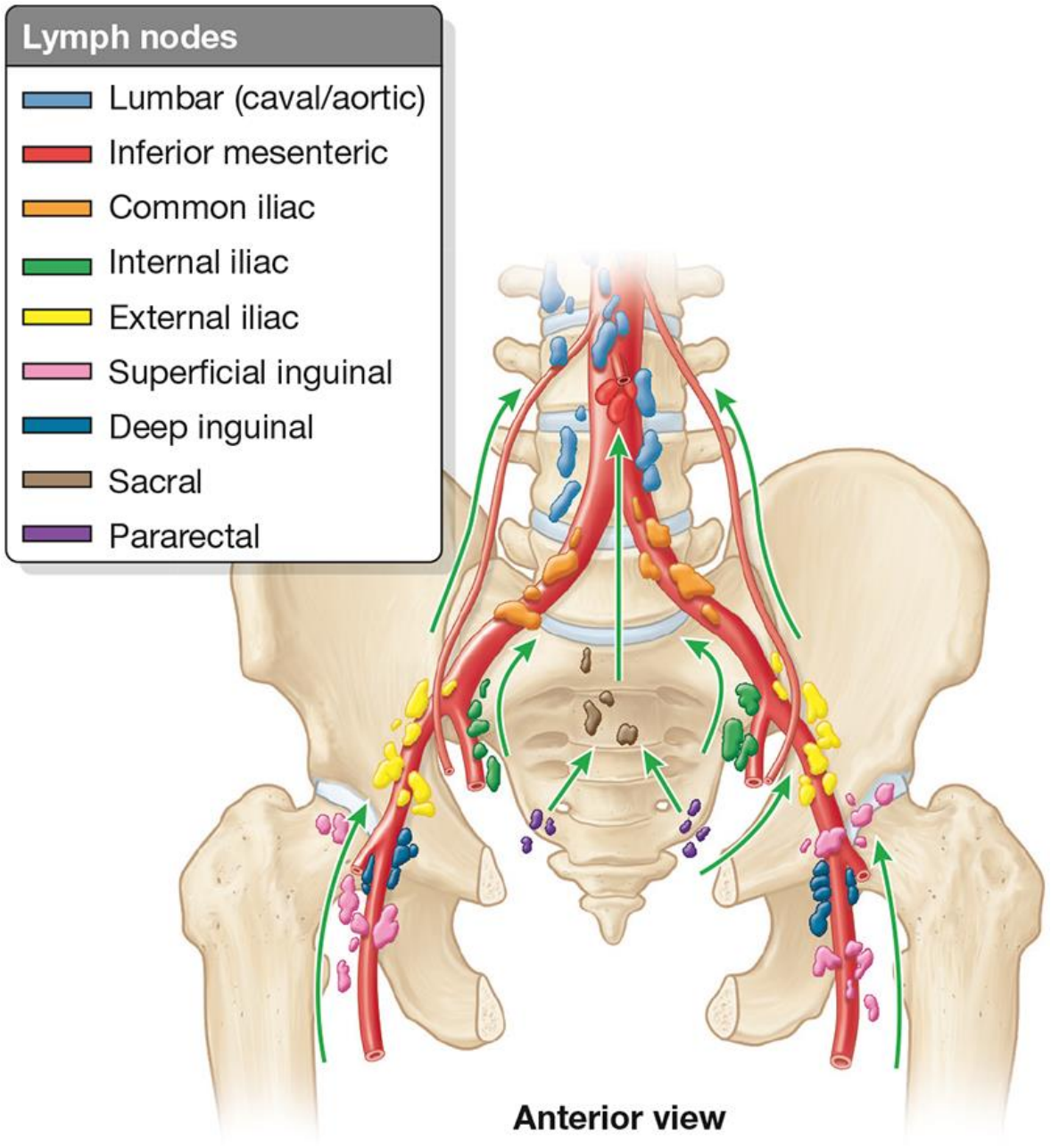
Peritoneal Reflections in Males (in Midline) in B
Peritoneum passes:
1 From the anterior abdominal wall
2 Onto apex of bladder, forming supravesical fossa
3 On the superior surface of the urinary bladder
4 2 cm inferiorly on the posterior surface of the urinary bladder
5 On the superior ends of the seminal glands
6 Posteriorly to line the rectovesical pouch
7 To cover the rectum
8 Posteriorly to become the sigmoid mesocolon

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Lymph Drainage of Pelvis and Perineum

- Pelvic nodes
  - Sacral
  - External iliac
  - Internal iliac
  - Common iliac
- Perineal drainage
  - External genitalia and skin to deep and superficial inguinal and internal iliac
  - Gonads to lumbar

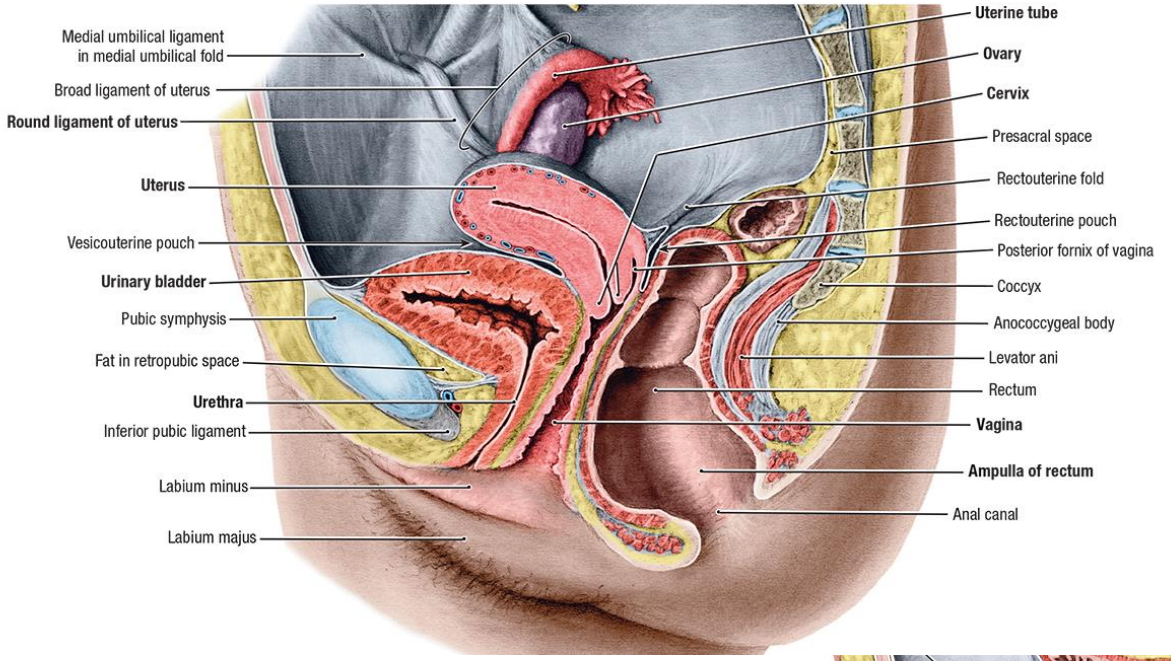


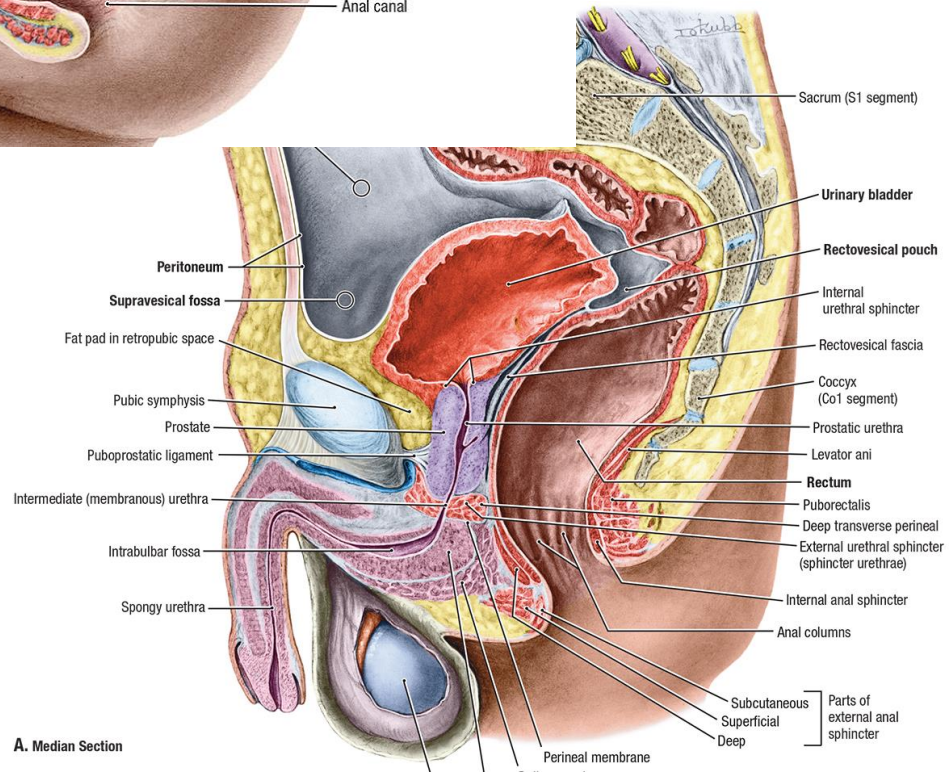
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Pelvic Viscera

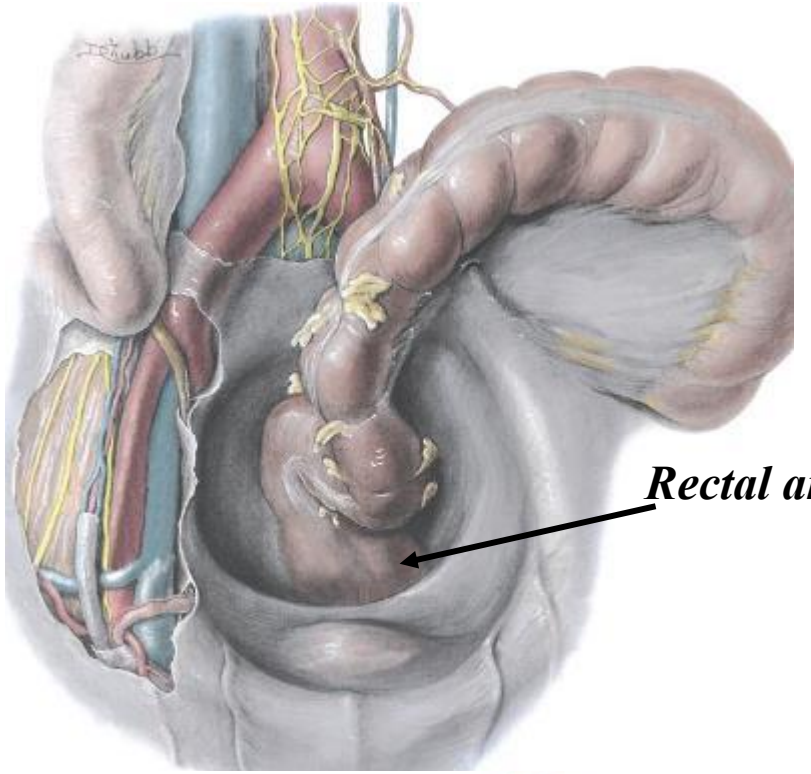
- Retro- or inferoperitoneal
- Inferior part of the GI tract
- Urinary Bladder
- Distal Ureters
- Reproductive systems

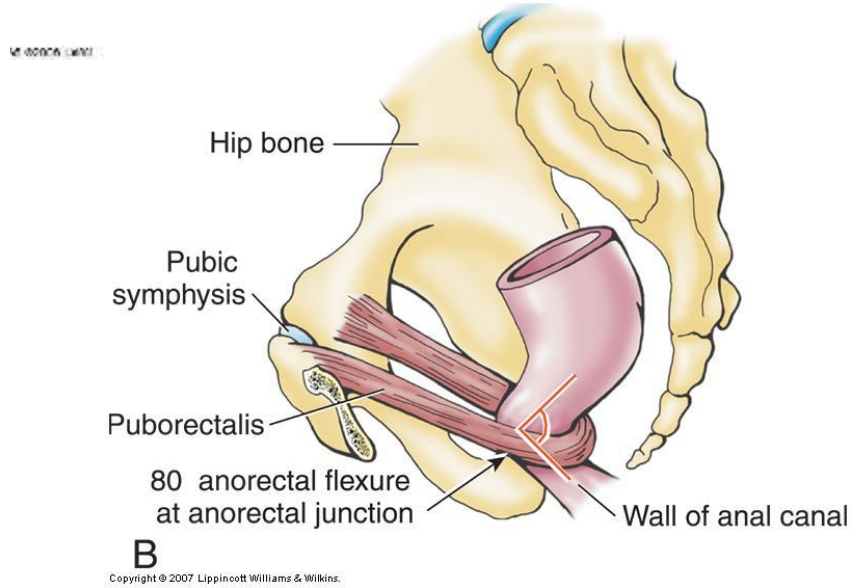




Rectum

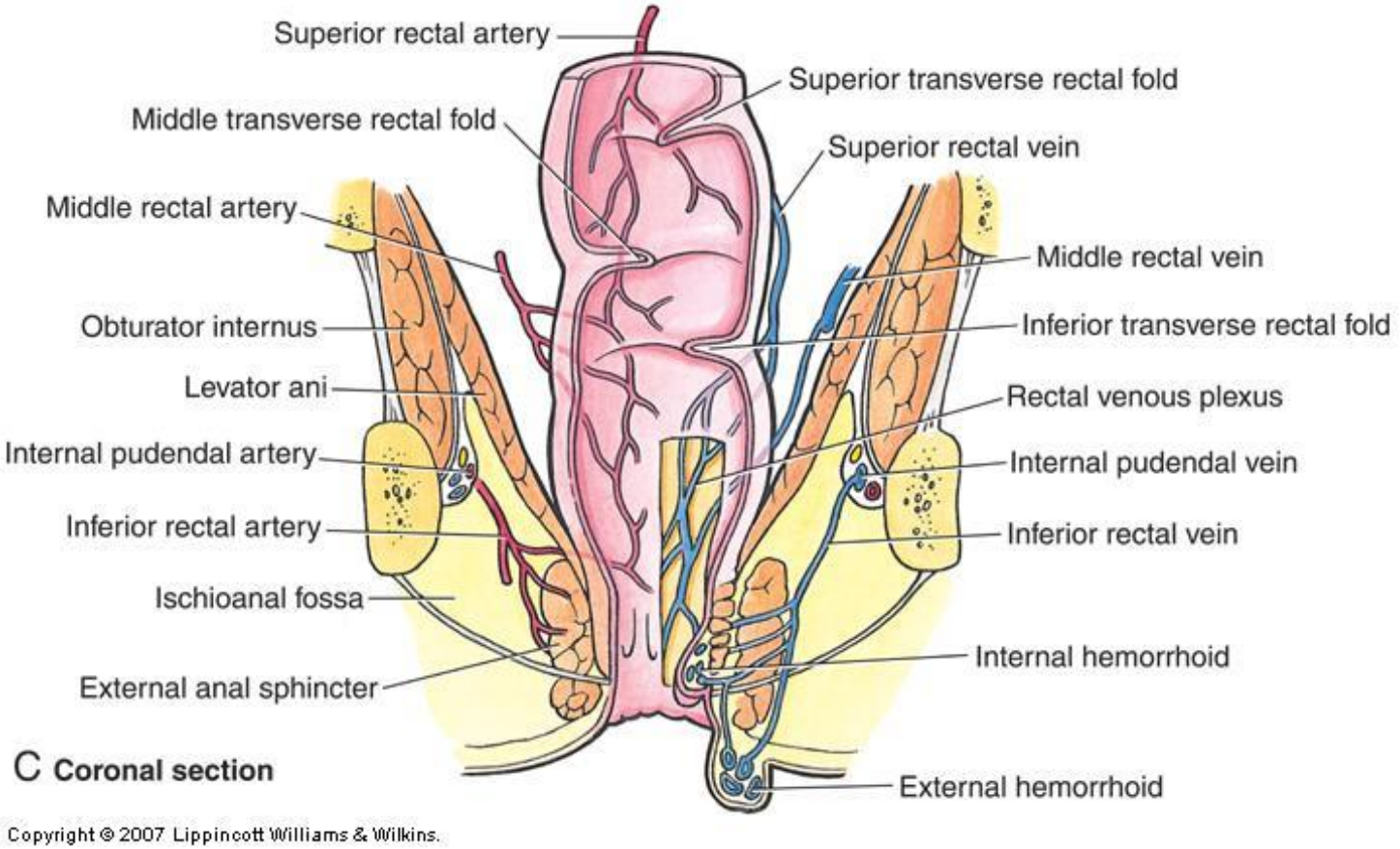
- Only part of the alimentary tract contained in lower pelvis
- Continuous with the sigmoid colon at the rectosigmoid junction (S3)
- Anorectal flexure aligns with the tip of the coccyx
  - Terminal region that turns sharply posterior before entering the anal canal
  - Roughly 80° angle
  - Anorectal flexure necessary for fecal continence
- Maintained by the tonus of the Puborectalis muscle
  - Sling around the rectum
  - Non-relaxing Puborectalis results in chronic constipation





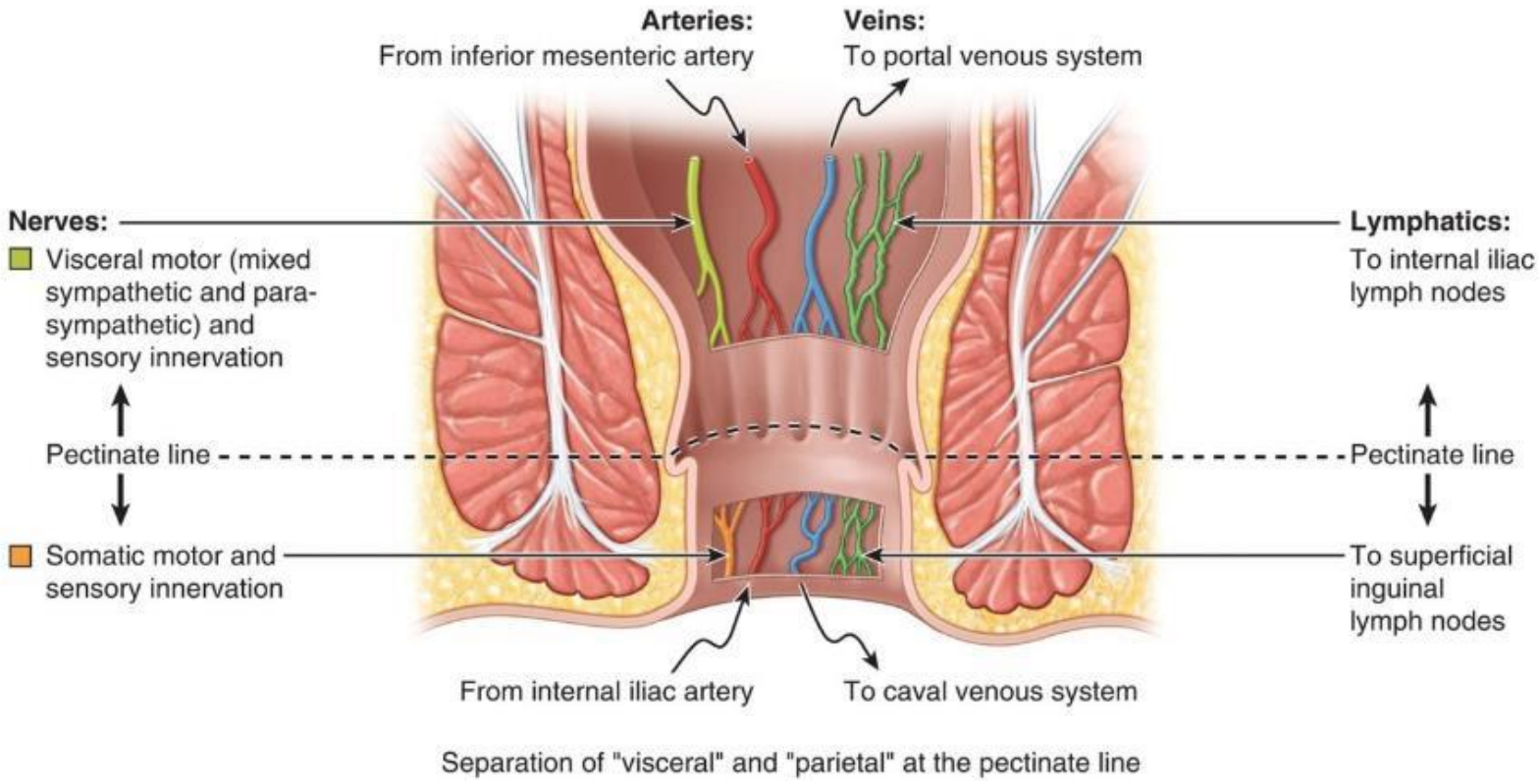
Rectum

- Internal folds also control fecal storage and continence
  - 3 Lateral flexures creating 3 sharp transverse rectal folds
- Vasculature
  - Inferior mesenteric artery via the Superior Rectal artery
  - Middle rectal arteries comes from the internal iliac/anterior branch
  - Inferior rectal come off the internal pudendal artery
- Venous return moves through the rectal venous plexuses
  - Internal below the epithelium
  - External deep to the muscular wall



Anal Canal

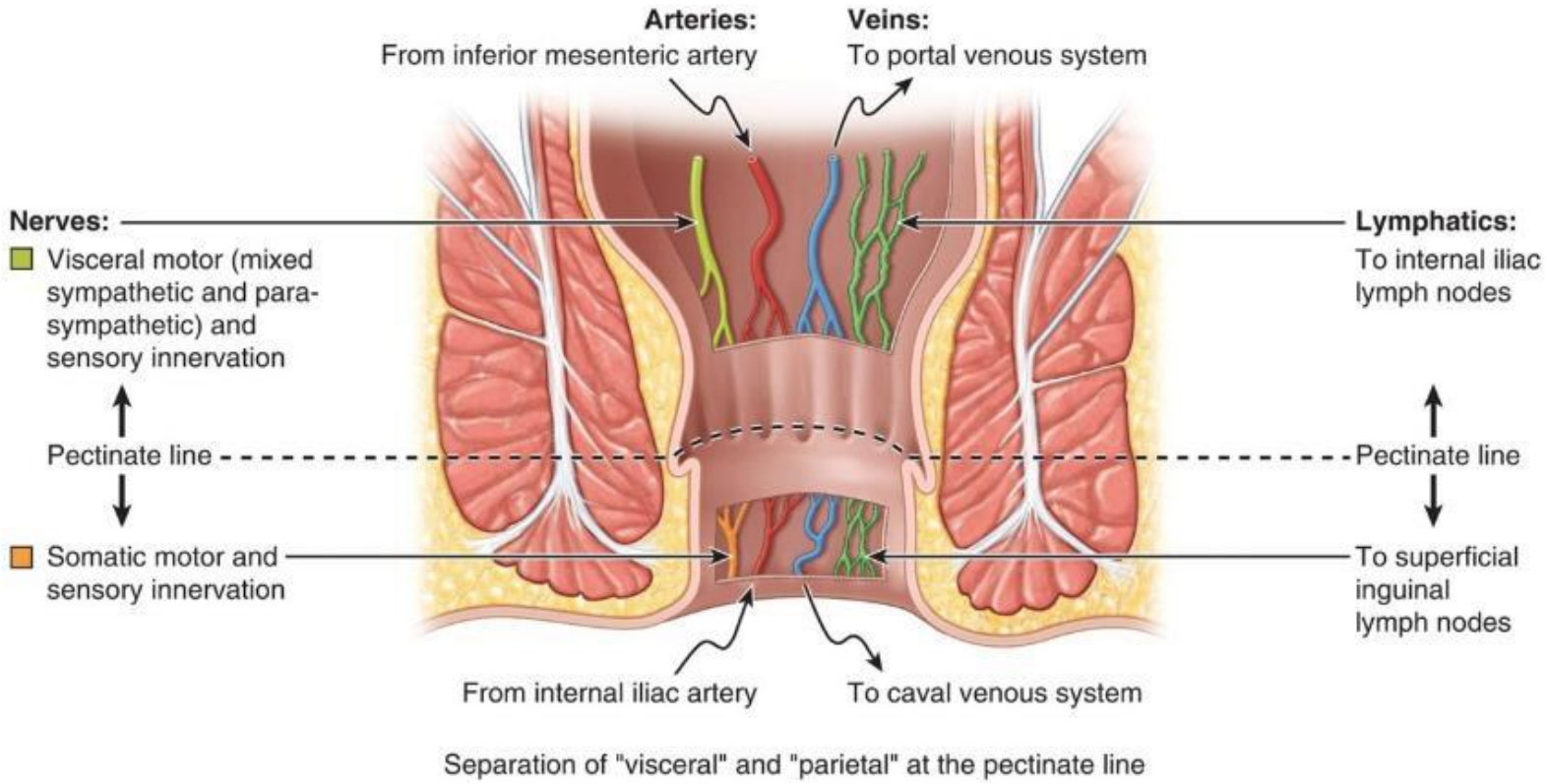
- Housed in perineum, ischioanal fossa
- Internal and External anal sphincters
  - Internal – Autonomic
  - External – Somatic
- Mucous membrane formed into anal columns
- Columns end in anal valves
  - Anal sinuses secrete mucus during defecation
- Pectinate line is serrated edge formed by valves
  - Delineates regions of anal canal



Separation of "visceral" and "parietal" at the pectinate line

Internal Anal Canal Neurovasculature

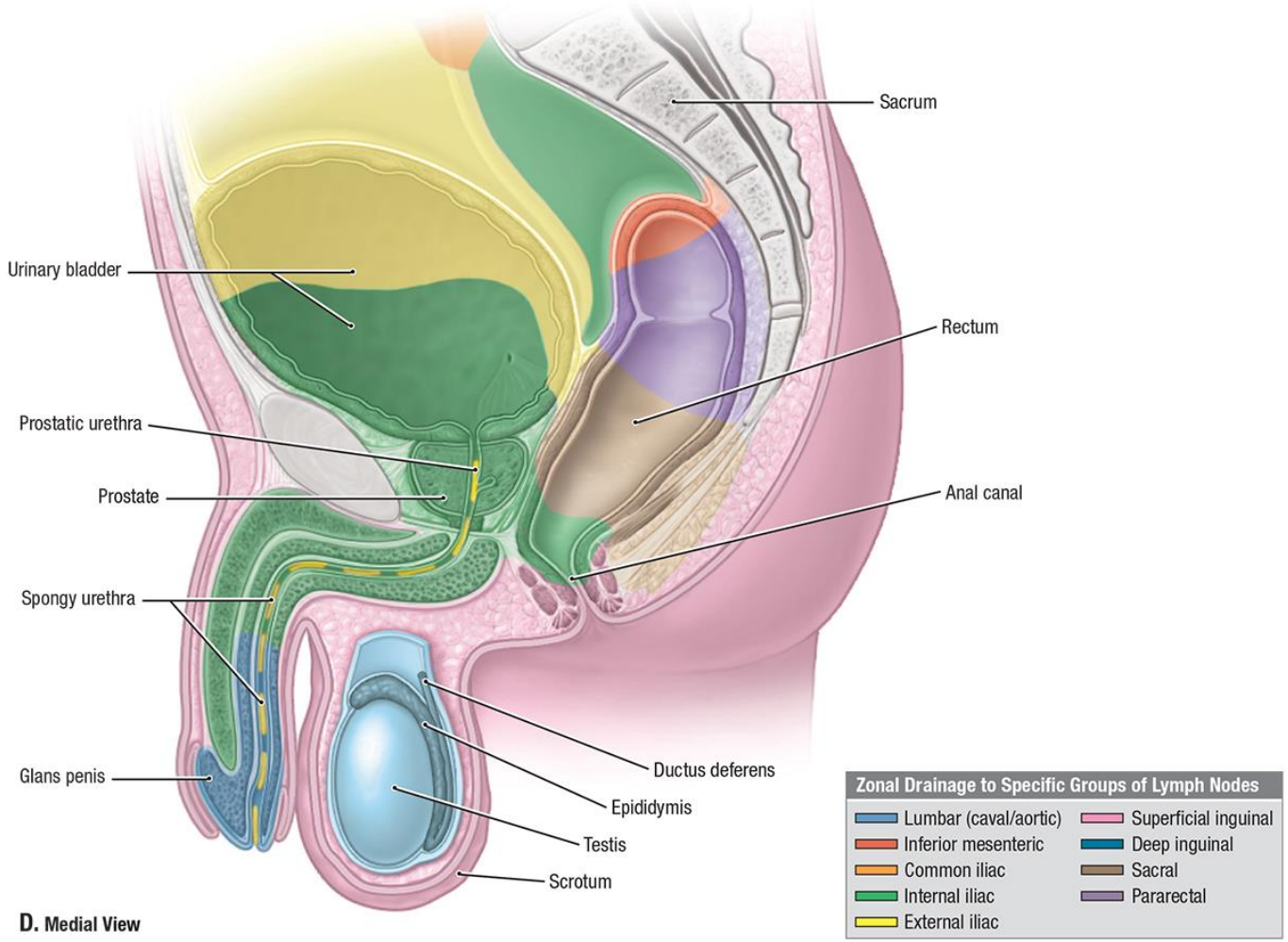
- Superior Anal Canal is visceral
  - From hindgut/endoderm developmentally
  - Innervation is autonomic
  - Blood from inferior mesenteric via superior rectal; Back to portal system
  - Lymph to internal iliac
- Inferior Anal Canal is somatic
  - Embryonic ectoderm
  - Somatic innervation from inferior rectal nerve off pudendal
  - Blood from internal iliac via inferior rectal to vena cava
  - Lymph to superficial nodes



Separation of "visceral" and "parietal" at the pectinate line

Lymphatic Drainage Rectum and Anal Canal

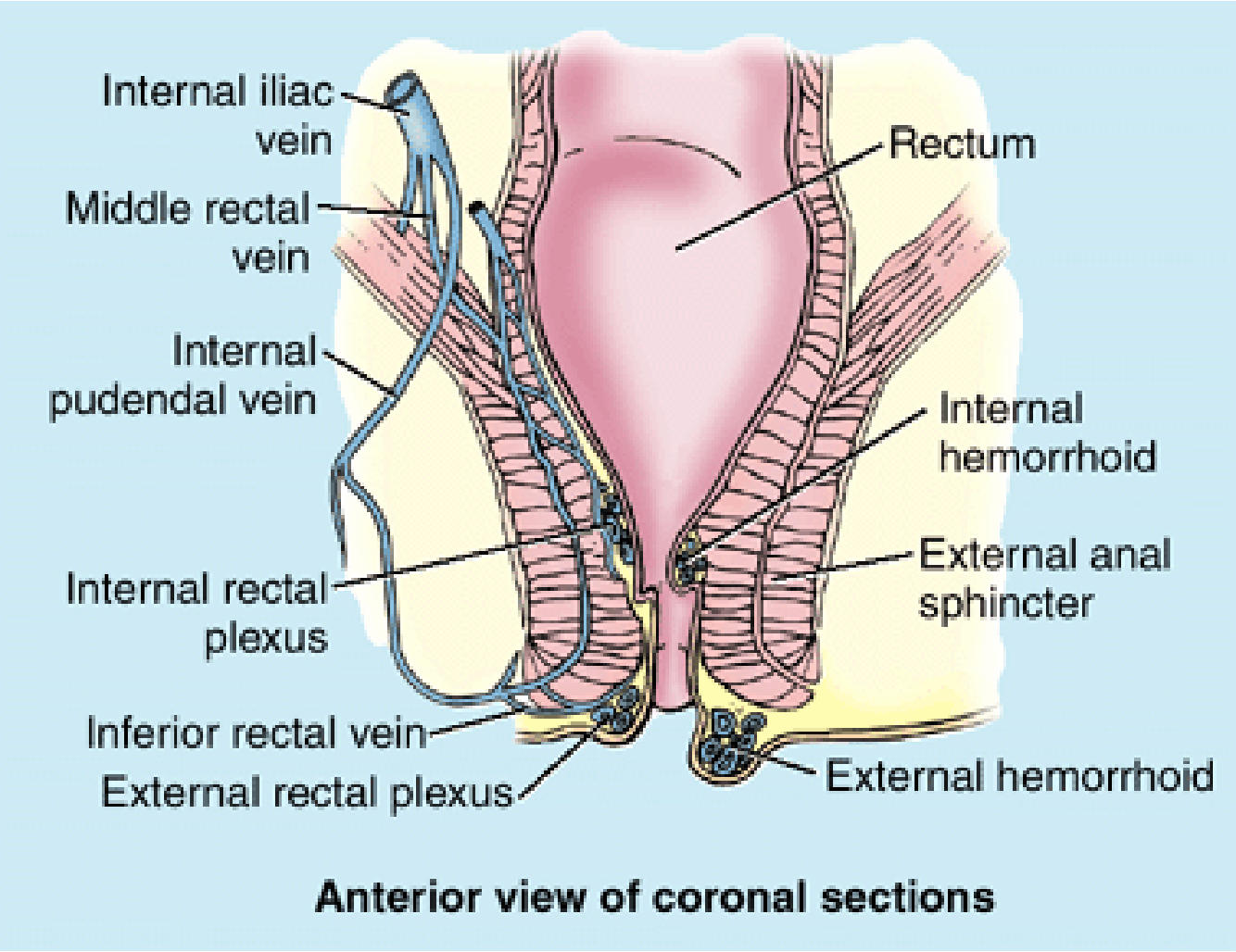
- No sex differences
- Distal anal canal to Superficial inguinal
- Distal rectum to Internal Iliac
- Proximal rectum to Sacral and Pararectal



Clinical

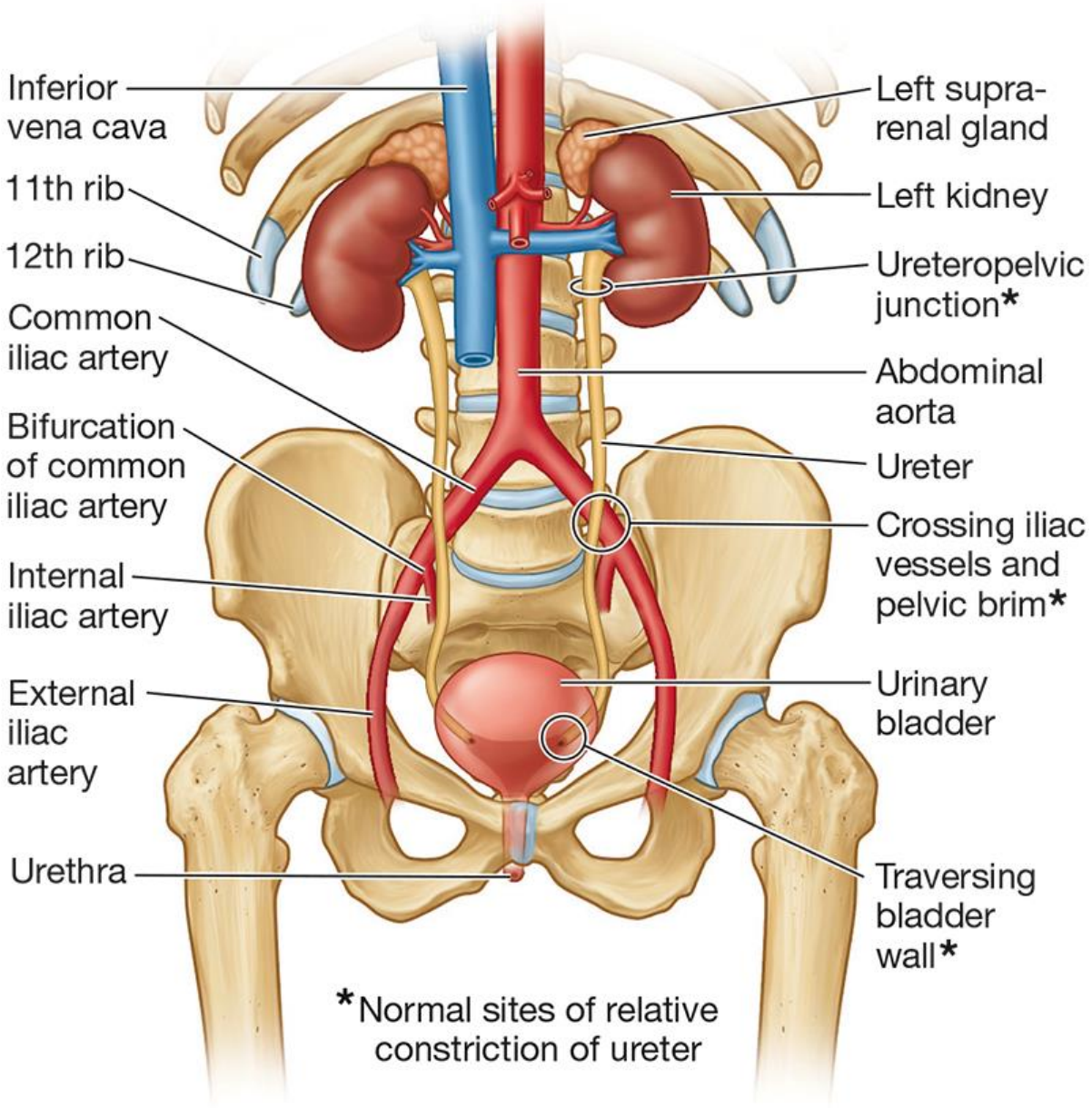
Hemorrhoids

- Internal
  - Prolapsed mucosa because of dilated internal venous plexus
  - Compressed by sphincters
    - Strangulation and ulceration
  - Bright red blood in stool
  - No pain/above pectinate line
- External
  - Clot in the external rectal plexus
  - Caused by impediments to normal venous return
    - Constipation, Intra-abdominal pressure
  - Very painful



Ureters

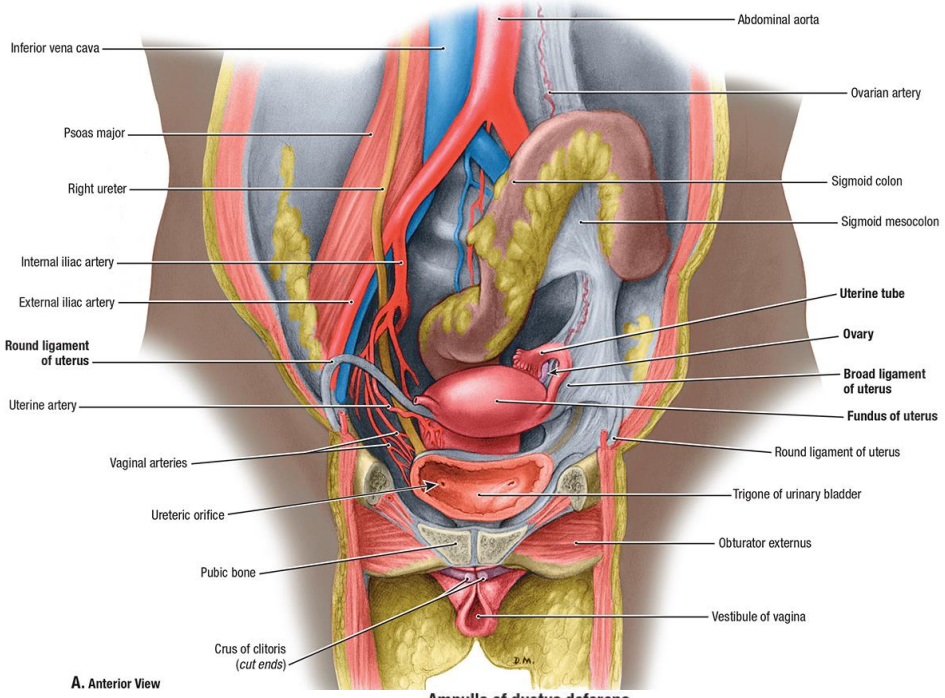
- Mostly retroperitoneal
- Areas of natural strictures
  - Renal pelvis
  - Crossing the common iliac
  - Enter the bladder
  - Stones can get caught
- Entry is oblique and protected by a one-way valve
- Blood supply is “scavenged” from any artery passed
- Sensory mostly refers to T10 – L2

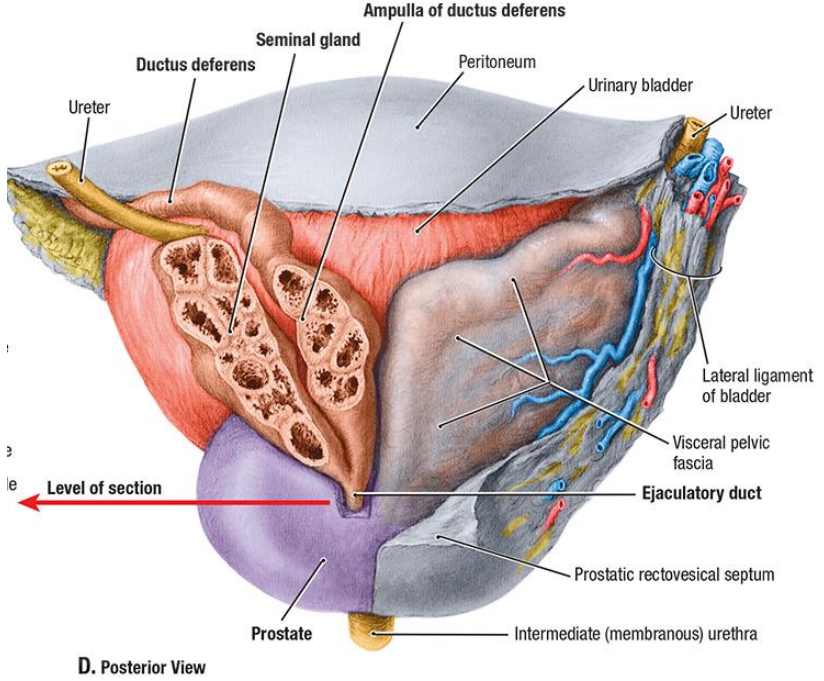


(A) Anterior view

Ureteric Relationships

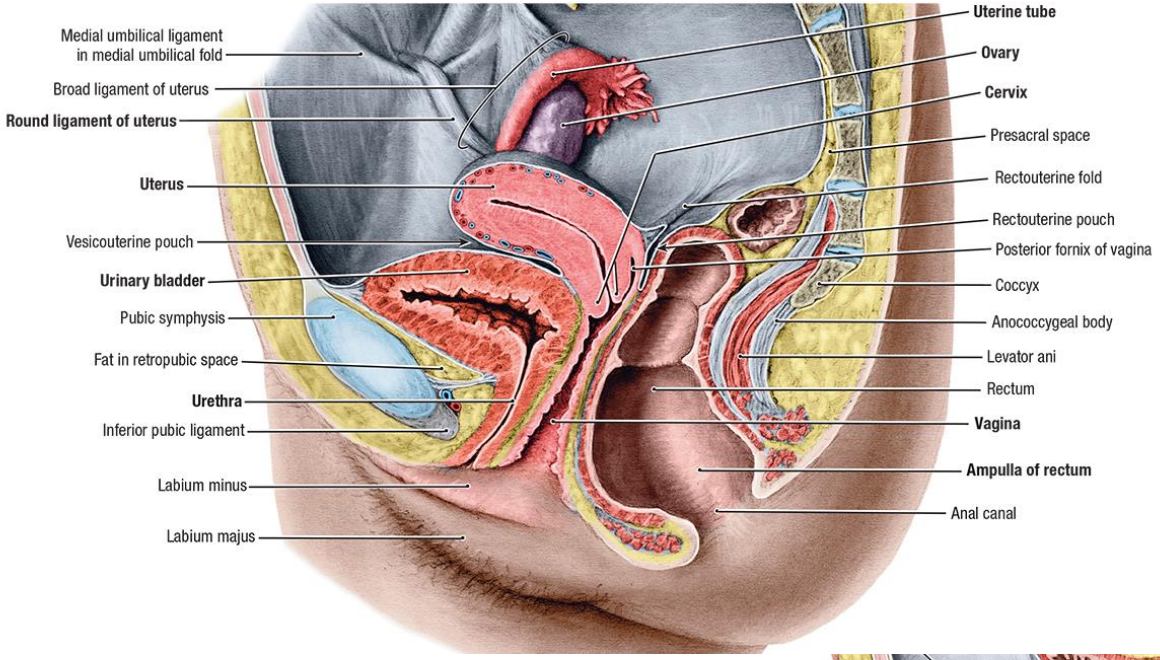
- Passes under the uterine artery (when present)
  - Must be ID'd and protected during hysterectomies
- Passes under ductus deferens on the posterior bladder wall

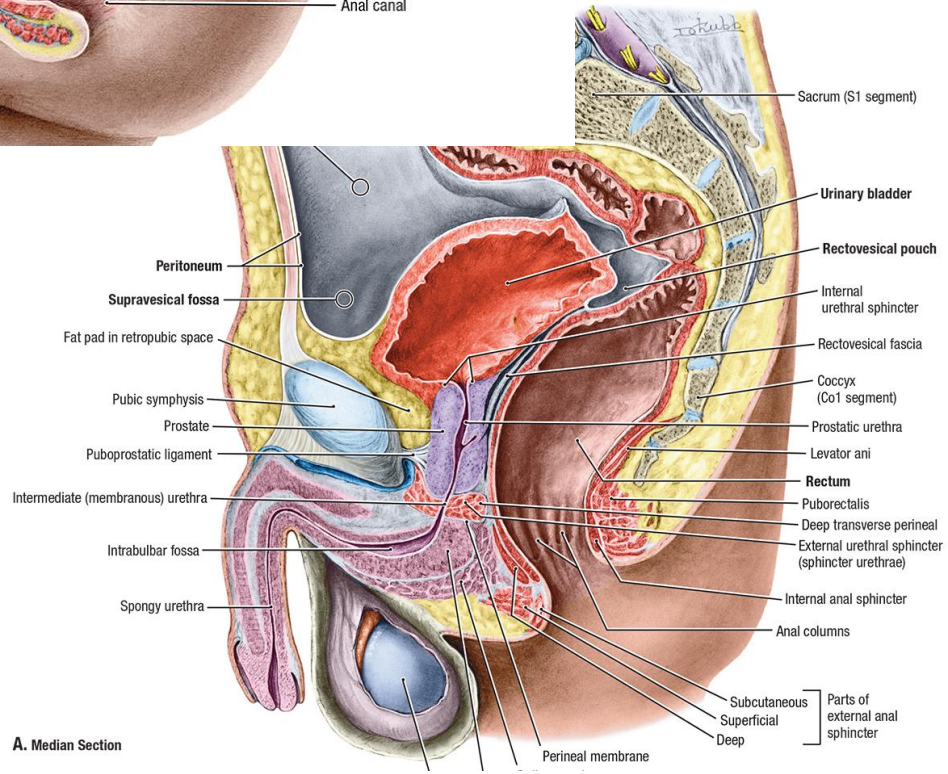




Urinary Bladder

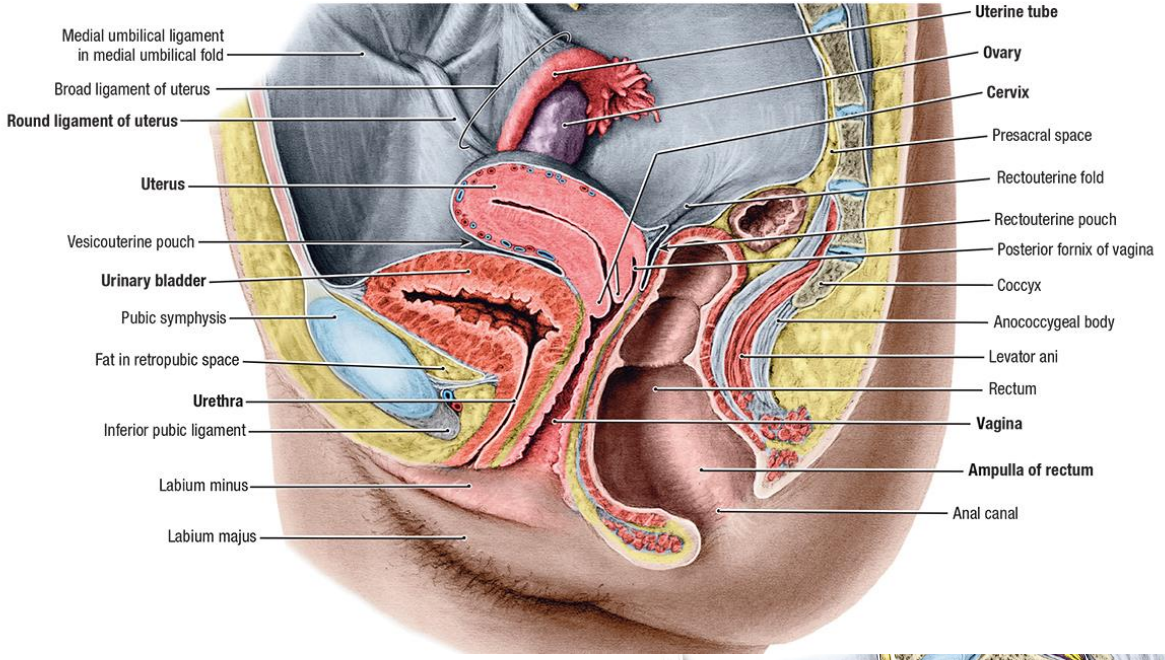
- Hollow organ
  - Strong Muscular Walls
  - Detrusor muscle
- Lies inferior to the peritoneum
- When empty rests fully in the lesser pelvis
  - When full it can extend up to the umbilicus
- Fairly mobile within the cavity
- Neck held in place by strong pubic ligaments
  - Puboprostatic ligament
  - Pubovesicular

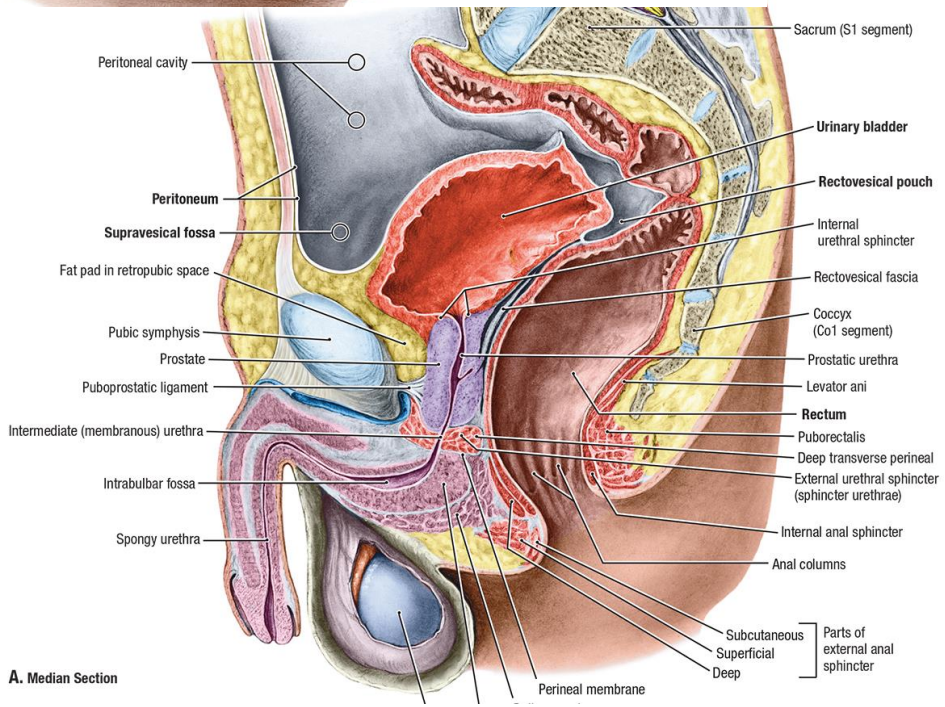




Urinary Bladder

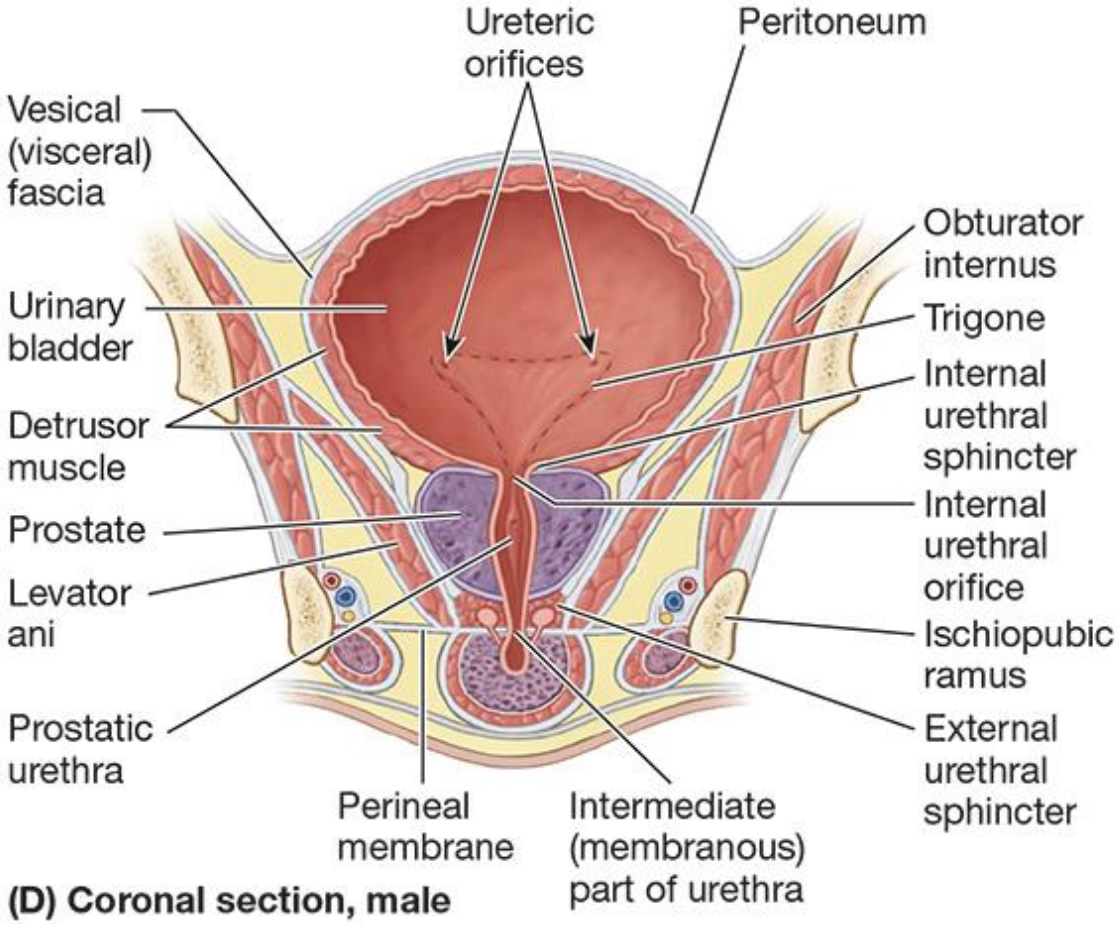
- Apex of the bladder projects anteriorly
- Fundus is opposite the apex
  - Near rectum or vagina
- Body is between the apex and fundus
- Divided into regions
  - Superior surface Covered by peritoneum
  - Two inferolateral surfaces
  - Posterior Surface
  - Fundus
  - Neck formed where Inferolateral surfaces and Fundus converge





Internal Features Urinary Bladder

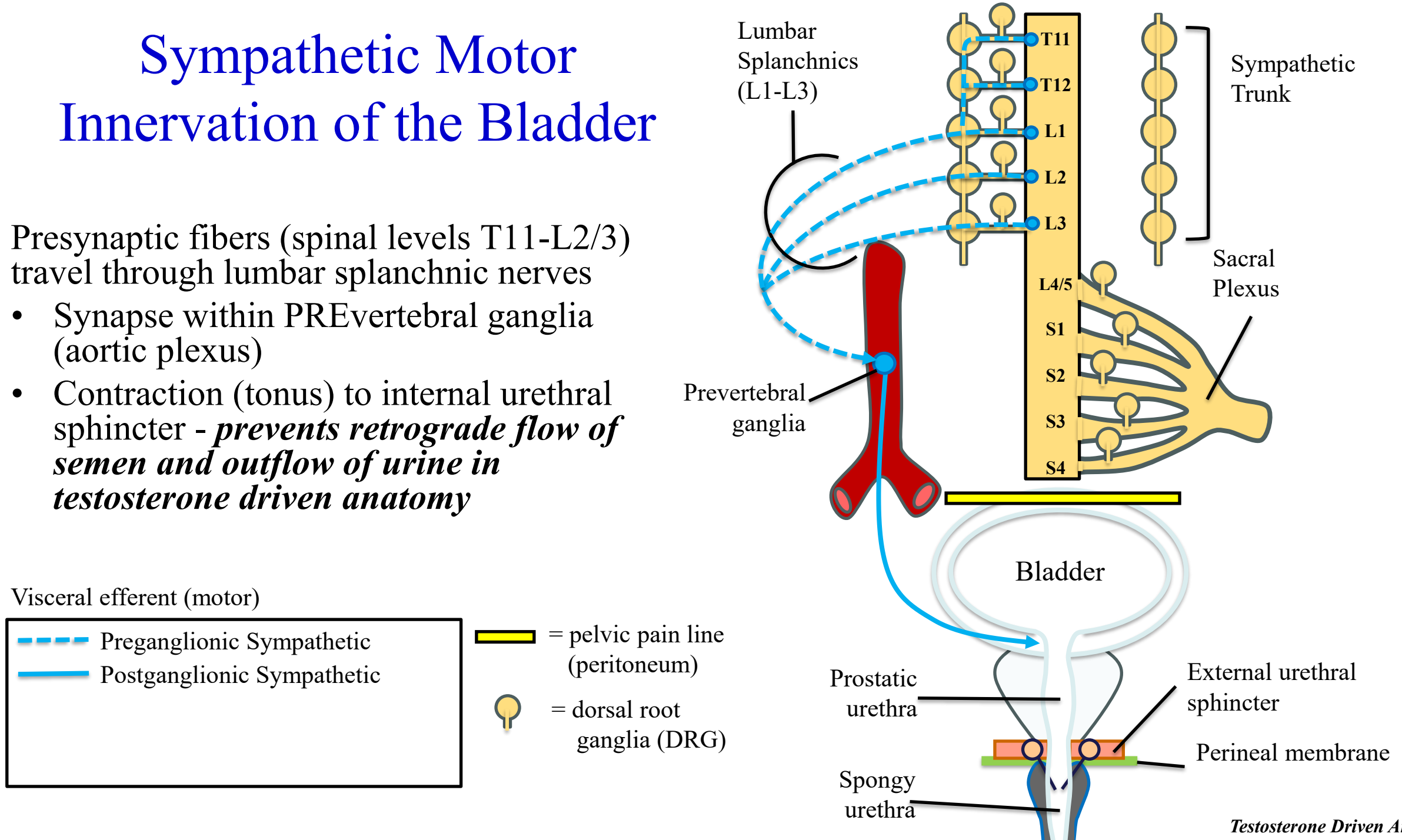
- Urine enters the bladder through the ureteric orifices
- Trigone
  - Triangular region of bladder wall
  - Smooth
  - Points formed by the ureteric orifices and the internal urethral orifice
- Internal urethral sphincter muscle
  - Testosterone makes larger
  - Involuntary smooth muscle
  - Closes during ejaculation to prevent urine release



Sympathetic Motor Innervation of the Bladder

Presynaptic fibers (spinal levels T11-L2/3) travel through lumbar splanchnic nerves

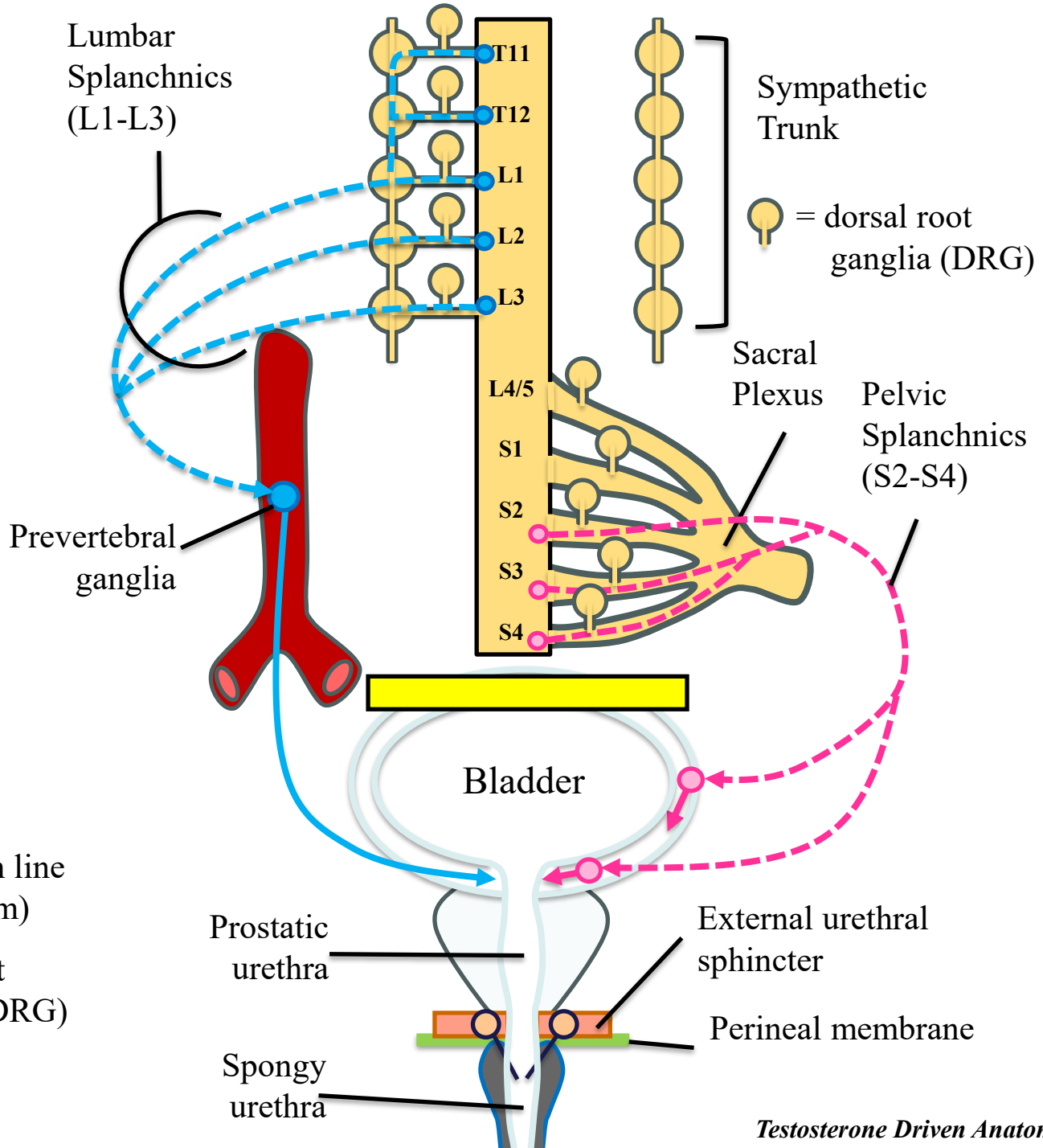
- Synapse within PREvertebral ganglia (aortic plexus)
- Contraction (tonus) to internal urethral sphincter - ***prevents retrograde flow of semen and outflow of urine in testosterone driven anatomy***



Parasympathetic Motor Innervation of the Bladder

Presynaptic fibers (spinal levels S2-S4) travel through pelvic splanchnic nerves

- Synapse within wall of the bladder
- Contracts detrusor (smooth) muscle and inhibitory to (relaxes) internal urethral sphincter (the neck of the bladder)- ***micturition***



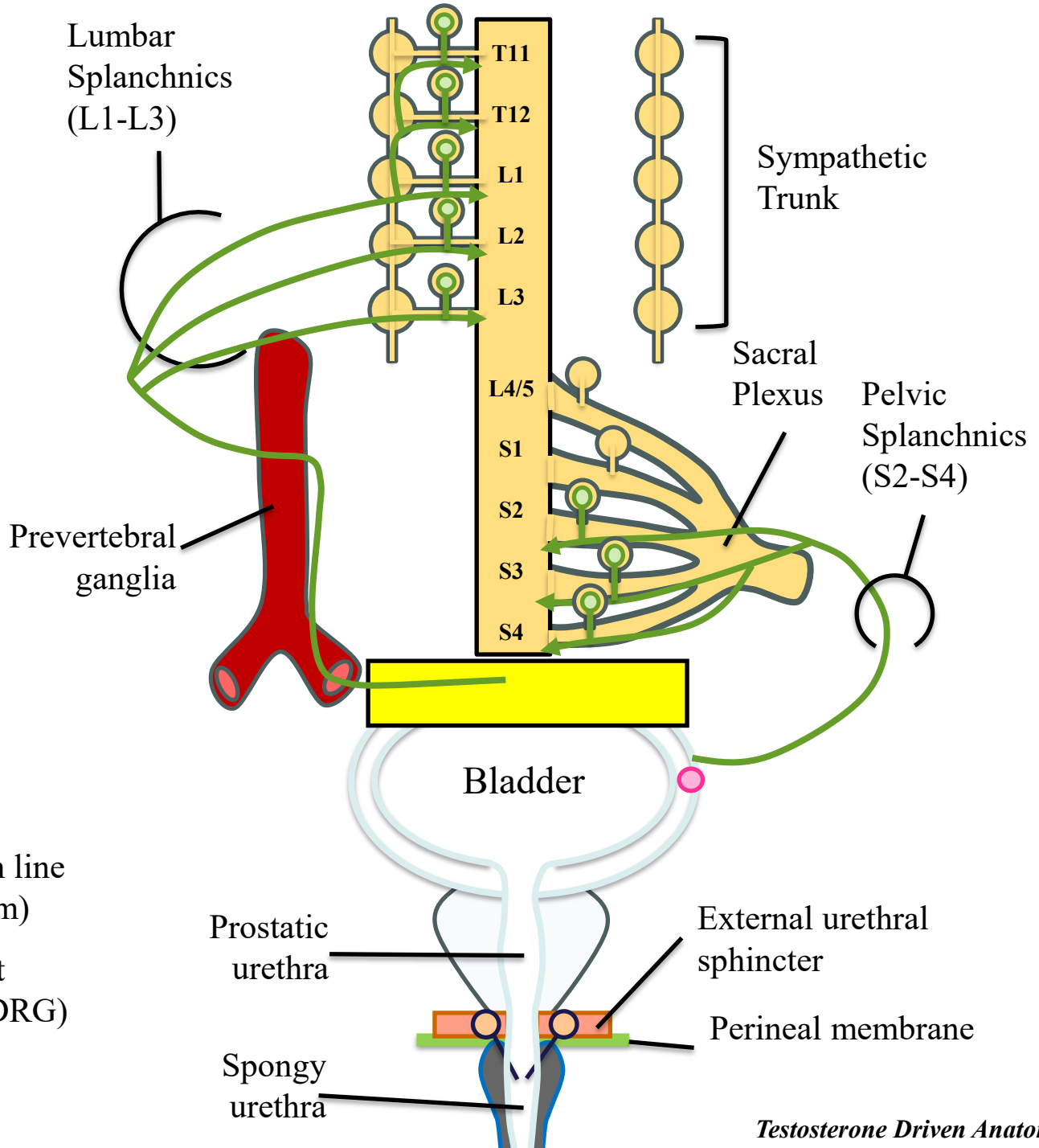
Visceral Afferent (Sensory) Innervation of the Bladder

Inferior to pelvic pain line

- *Body, inferolateral surfaces, fundus, neck, apex*
- ***Parasympathetic*** fibers to S2-S4 spinal ganglia

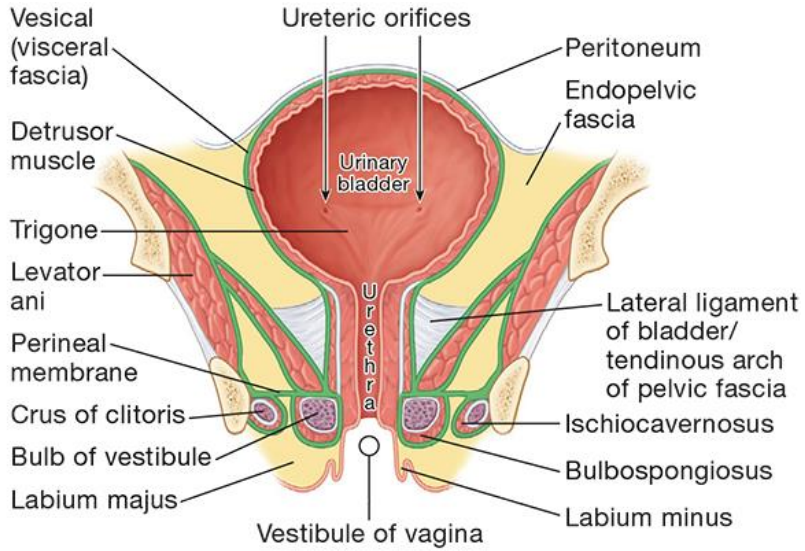
Superior to pelvic pain line

- Roof of the bladder
- ***Sympathetic*** fibers to T11-L2/L3 spinal ganglia

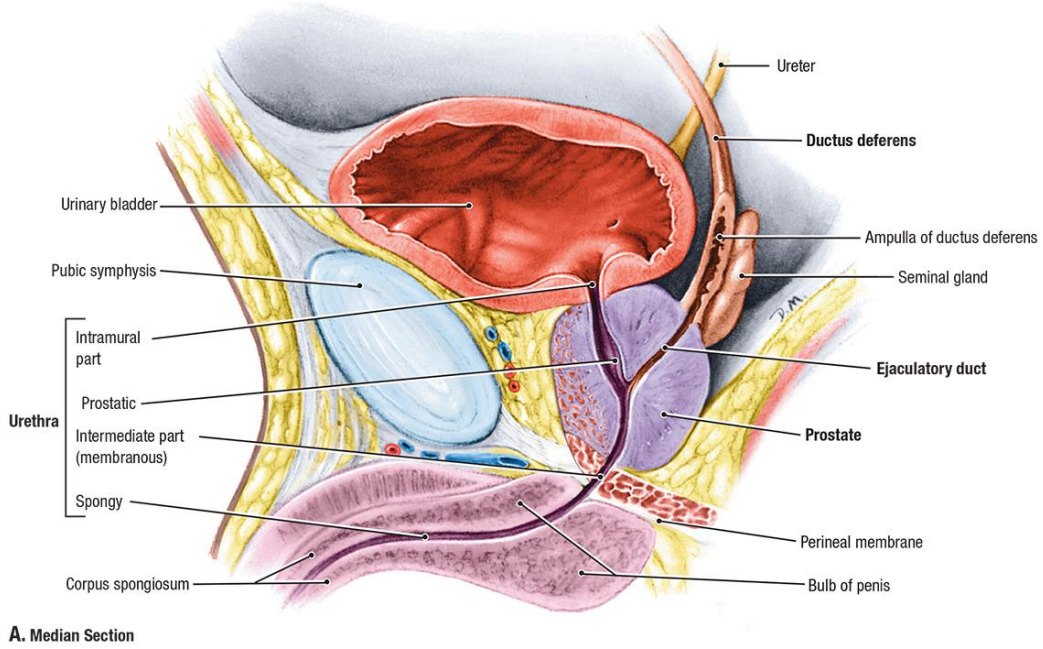


Urethra

- Sex differences
- Estrogen
  - Short
  - Opens into the vaginal vestibule
  - Not defined internal urethral sphincter
- Testosterone
  - Urethra also for sperm transmission
  - 4 regions
    - Intramural – through bladder wall
    - Prostatic – through prostate gland
    - Membranous – levator ani/perineal membrane
    - Spongy – corpus spongiosum



(C) Coronal section, female

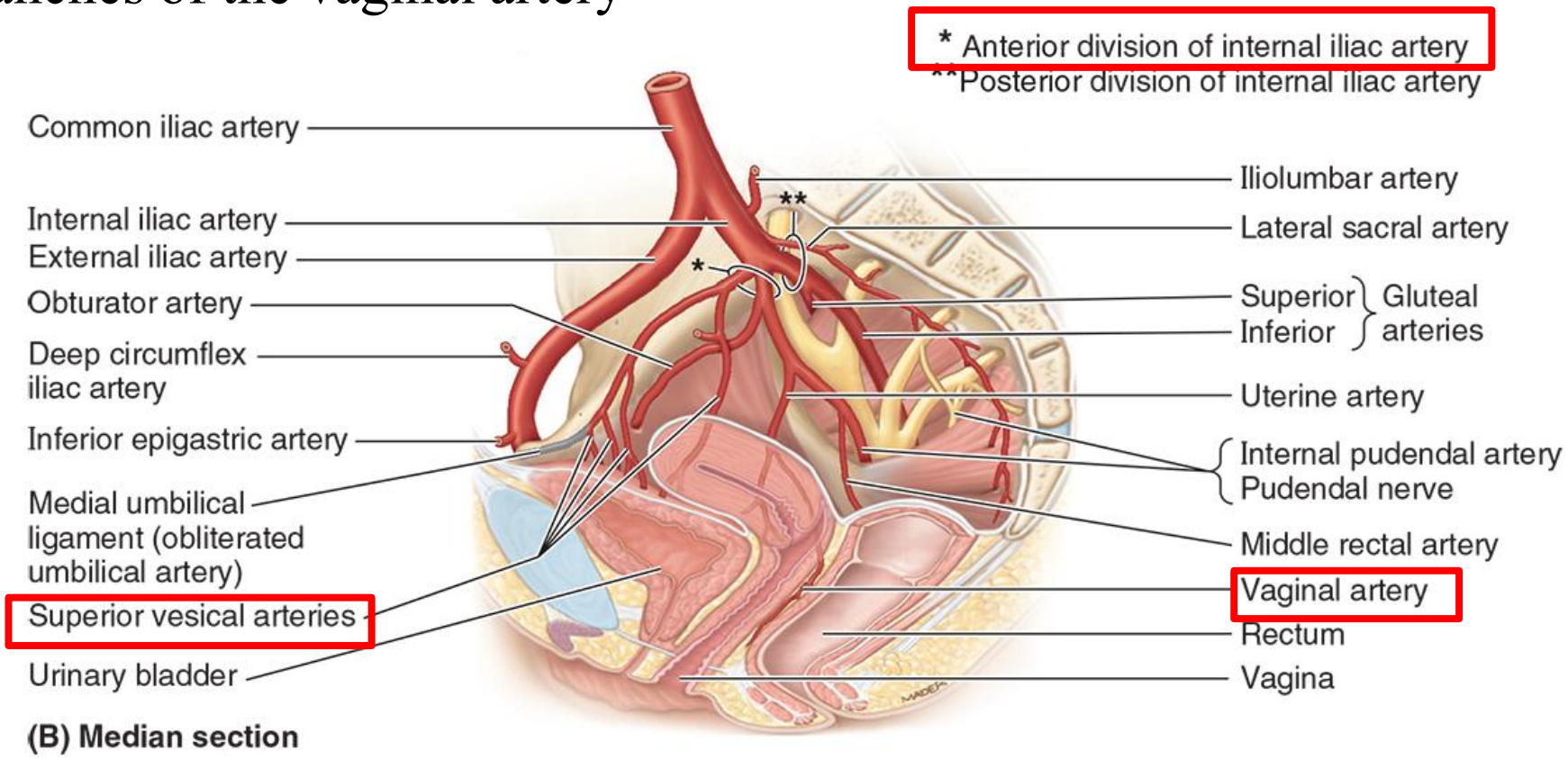


A. Median Section

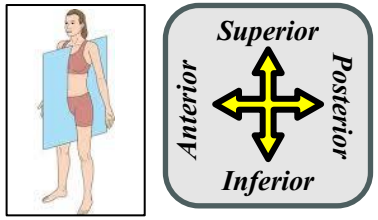
Arterial Supply of the Bladder and urethra

Branches of ***internal iliac (anterior division)***

- Superior surface - superior vesical arteries
- Inferolateral surface, fundus, neck of bladder – inferior vesicle artery (testosterone driven anatomy) and branches of the vaginal artery



Sagittal cut, medial view
Estrogen Driven Anatomy



Estrogen Driven Anatomy, Moore's, 9th ed., Fig. 6.17

Lymphatic Drainage Bladder and Urethra

- Bladder to Internal and External Iliac
- Distal urethra in Vaginal vestibule to Superficial Inguinal
- Prostatic and spongy urethra draining is mixed
  - External and Internal iliac
  - Deep inguinal

